

Between Violence and Politics: An Analysis of Global Terrorism

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Figure 1: Global Terrorism

Abstract

Terrorism has become a major global concern in recent decades, significantly impacting the stability of countries and regions, while raising complex questions about its underlying causes and patterns [26] [1]. Understanding how terrorist groups operate, how attacks evolve over time, and how political contexts influence such activity is essential for gaining deeper insights into this phenomenon. In this work, we adopt a data-driven approach to study terrorism using the Global Terrorism Database (GTD), complemented by political indicators used in the Varieties of Democracy Project (V-Dem) dataset. We explore the lifecycle of terrorist groups, examine how attack characteristics influence lethality and success, and analyse the role of political regimes in shaping terrorist activity. By leveraging interactive visualisations and dashboards, we transform complex data into intuitive and exploratory insights, enabling users to uncover patterns and better understand the dynamics of terrorism and its connection to political environments.

Keywords

Terrorism, Global, Analysis, Tableau

1 Introduction

Terrorism has become one of the most pressing and complex challenges of the modern world. Often characterized by the use of one-sided, indiscriminate violence against unsuspecting and unarmed victims, it raises fundamental questions about its underlying motivations and strategic rationale [9]. Beyond the immediate physical destruction, terrorism aims to generate fear, intimidation, and coercion, typically in pursuit of political, ideological, or social objectives. This deliberate targeting of civilians and non-combatants,

frequently in disregard of legal and moral constraints, contributes to its deeply controversial and multifaceted nature [9]. Over the past few decades, terrorism has significantly increased in scale and impact, posing a serious threat to the stability of countries and entire regions [26]. Its effects are long-lasting, often leaving societies in states of uncertainty and insecurity. At the same time, the concept itself remains contested, as different actors define and interpret terrorism according to political, social, and ethical perspectives [1]. Understanding terrorism is further complicated by its strong connection to political contexts. Some perspectives interpret it as a reaction to perceived injustices, ineffective governance, or failed negotiations between states and non-state actors [1], while others emphasize its strategic and organized nature. These differing viewpoints highlight the importance of studying terrorism not only as a set of isolated incidents, but as a phenomenon embedded in broader socio-political dynamics. In this work, we adopt a data-driven approach to explore patterns and characteristics of terrorism using the Global Terrorism Database (GTD), complemented by a dataset containing political indicators in the Varieties of Democracy Project (V-Dem) dataset. Before conducting any analysis, we carried out a review of existing literature in order to better understand the phenomenon of terrorism, its driving factors, and its relationship with political environments. This step was essential to provide context for the data and to guide the direction of our study. Following this, we formulate a set of research questions aimed at exploring different dimensions of terrorism, including group behaviour, attack characteristics, and the influence of political regimes. We then proceed with an exploratory analysis of the datasets to understand their structure, variables, and limitations, which informs the selection of appropriate methodologies. Finally, we apply a range

of analytical and visualisation techniques to uncover patterns and relationships within the data. The goal is not only to identify meaningful insights, but also to provide clear and interpretable visual representations that support the understanding of terrorism and its underlying dynamics.

2 Research Questions

In order to better understand the dynamics and patterns of terrorism, an initial step is to explore how terrorist groups operate and evolve over time. By analysing the characteristics of attacks and group behaviour, we aim to identify recurring patterns and key factors that define terrorist activity. With this in mind, we present the following questions:

- (1) What is the lifecycle of a terrorist group?
 - When do terrorist groups begin and cease their activity?
 - What types of attacks are most commonly associated with each group?
 - Which regions are most frequently targeted?
 - What are the typical levels of lethality (e.g., number of casualties)?
 - What types of weapons are used?

The objective of this set of questions is to characterize terrorist groups in a comprehensive way, identifying behavioural patterns, operational preferences, and temporal dynamics that define their lifecycle. To further deepen our understanding of individual terrorist events, we then focus on how different attributes of attacks interact and influence their outcomes. This leads to the following question:

- (2) How do attack type, target type, and region jointly influence the lethality and success of terrorist attacks over time?
 - Which combinations of attack and target types are associated with higher lethality?
 - How does the success rate of attacks vary across regions?
 - Are there temporal trends in these relationships?

The objective here is to uncover relationships between key variables in the dataset and understand how they contribute to the impact and effectiveness of terrorist incidents. Finally, in order to place terrorism within a broader socio-political context, we incorporate political data to examine how different governance structures may influence terrorist activity. This leads to the following question:

- (3) How do political regimes shape the frequency, lethality, and success of terrorist attacks?
 - Are certain types of political regimes associated with higher frequencies of attacks?
 - Do regime characteristics influence the lethality of incidents?
 - Is there a relationship between political stability and attack success rates?

The objective of this final question is to explore the interaction between political environments and terrorism, identifying whether and how governance and political conditions may contribute to observable patterns in terrorist activity.

3 Dataset GTB

Global Terrorism Database is the main dataset for this project. Representing terrorist attacks around the world from 1970 to 2017 (except 1993). This dataset can be accessed in [25], downloading the .zip folder with 30Mb (162.81Mb extracted), in the format .csv. At the time, Version 3 was the one available. It contains 181692 rows, 181691 samples, and 135 variables/columns.

The Global Terrorism Database (GTD) includes both spatial and temporal variables with different levels of extent, granularity, and resolution. The spatial extent of the dataset is global, covering terrorist events across multiple countries and regions. Variables such as `country_txt` and `natlty1_txt` operate at the country or nationality level with nominal resolution, while `region_txt` aggregates data at a broader regional scale. The variable `city` provides a finer spatial granularity at the city level, also with nominal resolution. In contrast, latitude and longitude offer the highest spatial granularity, representing exact point locations with continuous resolution. Regarding temporal dimensions, the dataset spans from 1970 to the present, providing a global temporal extent. The variable `year` captures time at a yearly granularity with discrete resolution, while `imonth` and `iday` refine this to monthly and daily levels, respectively, also with discrete resolution. The variable `extended` represents event duration, distinguishing whether an attack lasted more than 24 hours, and is recorded with binary categorical resolution at the event level. Together, these variables allow for multi-scale spatial and temporal analysis of terrorist activity.

The most significant variables for this project are:

- `year`, `imonth` and `iday`
- `region`, `country` and `city`
- `groupname` - name of the terrorist group
- `attacktype` - type of attack
- `weaptype`, `weapsubtype` - type and subtype of weapon used in the attack
- `targettype`, `targetsubtype` and `nationalitytarget` - the target (person or place) and subtarget (specified) of the attack and the nationality
- `deaths` - number of people killed
- `injured` - number of people injured
- `success` - attack's success
- `claimed` and `claimmode` - if it was claimed by any group or person and with which mode

3.1 Index table about the most significant variables

Table 1: Dataset Variables Description

Variable	Original Name	Description	Type	Example
year	year	Year of the attack	int	2000
month	month	Month of the attack	int	3
day	day	Day of the attack	int	27
region	region_txt	Region where the attack occurred	string	Western Europe
country	country_txt	Country where the attack occurred	string	Mexico
city	city	City where the attack occurred	string	Seattle
groupname	gname	Name of the terrorist group	string	MANO-D
attacktype	attacktype1_txt	Attack's type	string	Bombing/ Explosion
weaptype	weaptype1_txt	Type of weapon used	string	Firearms
weapsubtype	weapsubtype1_txt	Subtype of weapon used	string	Molotov Cocktail
targettype	targettype1_txt	Target of the attack	string	Government (Diplomatic)
targetsubtype	targetsubtype1_txt	Specified subtarget	string	Aircraft (not at airport)
nationalitytarget	natlty1_txt	Nationality of the target	string	United States
deaths	nkill	Number of people killed	int	47
injured	nwound	Number of people injured	int	9
success	success	Whether the attack succeeded	binary	1
claimed	claimed	Whether the attack was claimed	binary	0
claimmode	claimmode_txt	Claim mode	string	Letter
eventid	eventid	Event id (year, month, day)	long	197001000002

3.2 EDA details and data cleaning

To clean the dataset, unimportant columns were dropped, difficult names were changed for more suggestive ones. Rows with countries that are not present in the other dataset were also dropped, turning from 181691 samples to 170225. Blank/Null spaces were left without changes since Tableau is capable of handling that. Assigning each variable to its corresponding field (dimension and metric) was also a necessity. In the end, hierarchies were created in spatial data: region - country - city, in temporal data: year - month - day, and in other attributes: targettype - targetsubtype and weaptype - weapsubtype. In order to understand better data, the following questions were tried to answer.

- (1) Checking the distribution of key variables (e.g. how many attacks per year?)

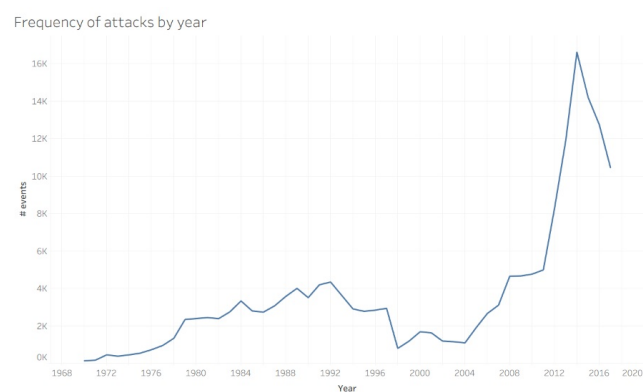


Figure 2: Frequency of attacks per year

The time series reveals three distinct eras. From 1970 to the early 1990s, attacks grew gradually, peaking around 4,500 in 1992. The

mid-1990s saw a sharp drop. The post-2001 era tells a different story entirely: a dramatic and sustained surge that culminated in a record ~16,500 attacks in 2014, with a sharp drop right after. The trend by 2017–18 still sits far above historical norms (~10,500), suggesting terrorism remained elevated compared to any pre-2010 period.

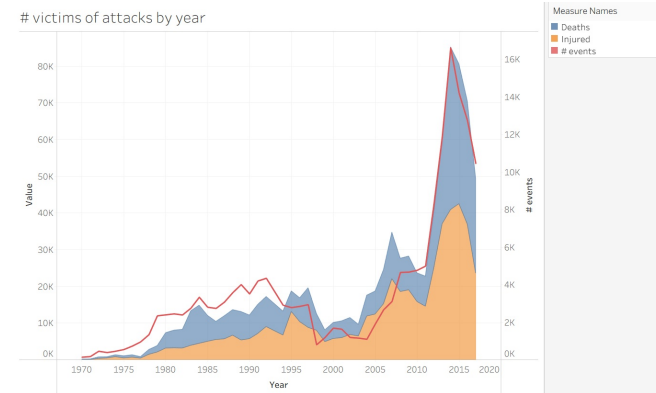


Figure 3: Victims of attacks per year

The number of victims, both deaths and injuries, closely tracks the frequency of attacks over time. However, there is a notable period in which the number of events declined while the injured count remained elevated, suggesting more destructive individual attacks during that window. Around 2014–2015, both attack frequency and victim counts reached their peak, with an exponential spike followed by a sharp decline in subsequent years. This peak likely reflects both a genuine surge in activity (driven largely by ISIL, as it can be observed forward in these paper) and improved global reporting practices, which may have inflated recorded figures relative to earlier decades.

- (2) Looking at which countries/regions appear most

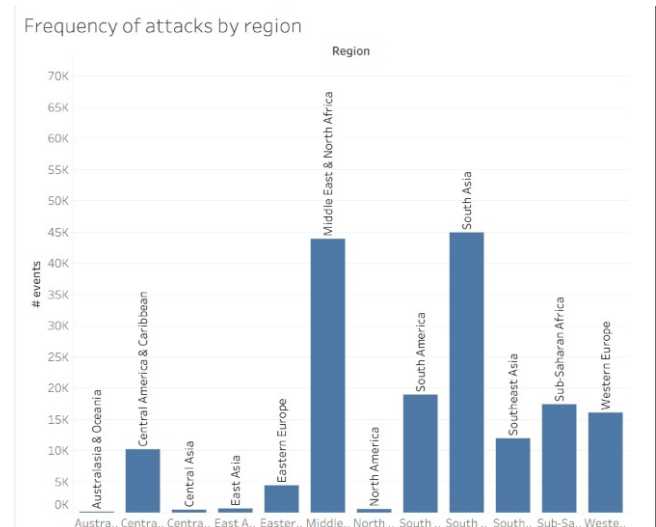


Figure 4: Frequency of attacks by region

South Asia (~45k) and Middle East and North Africa (~44k) are by far, the most affected regions, together accounting for roughly half of all recorded attacks globally. South America (~19k) and Sub-Saharan Africa (~17k) are the next most impacted. On the other hand, East Asia (~4k), North America (~1k) and Australasia and Oceania have very low counts relative to global figures, reflecting both geographic distance from active conflict zones and strong domestic security institutions.

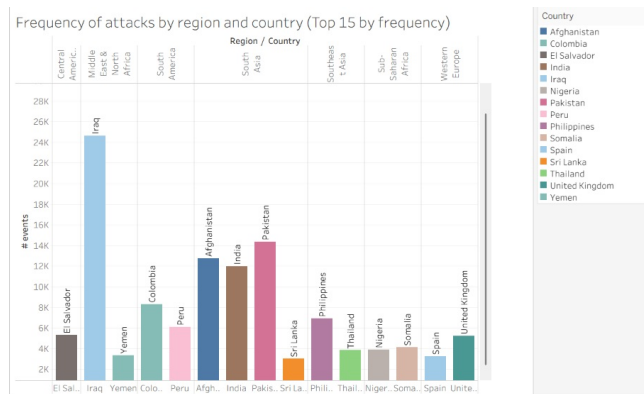


Figure 5: Frequency of attacks by region and country (Top 15)

Iraq (~25K) stands alone as the single most affected country in the dataset, nearly double that of the next highest, Pakistan (~15K) and Afghanistan (~13K), with India (~12K) coming fourth. In South America, Colombia (~8.5K) stands out. Followed by Philippines (~7k) in Southeast Asia.

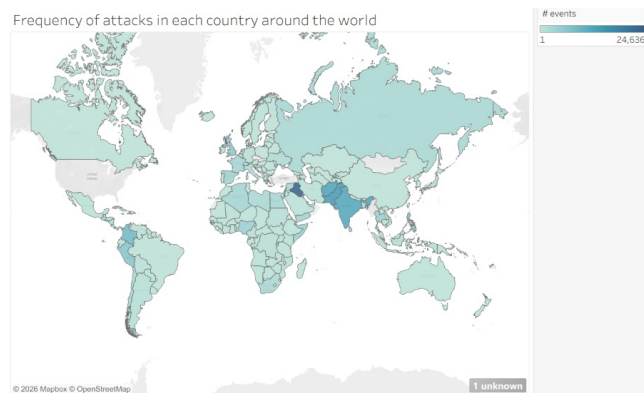


Figure 6: Frequency of attacks by country

At the country level, Iraq and Pakistan are by far the most affected nations, with Afghanistan, India, and Colombia also registering high event counts. The map visually confirms the broader regional trend: terrorism has become increasingly concentrated in the Middle East and South Asia, while regions that were historically more active, such as Western Europe and Central America, now show comparatively low attack frequencies. The distribution is heavily

skewed, with a small number of countries accounting for a disproportionate share of all recorded events.

Number of deaths by region

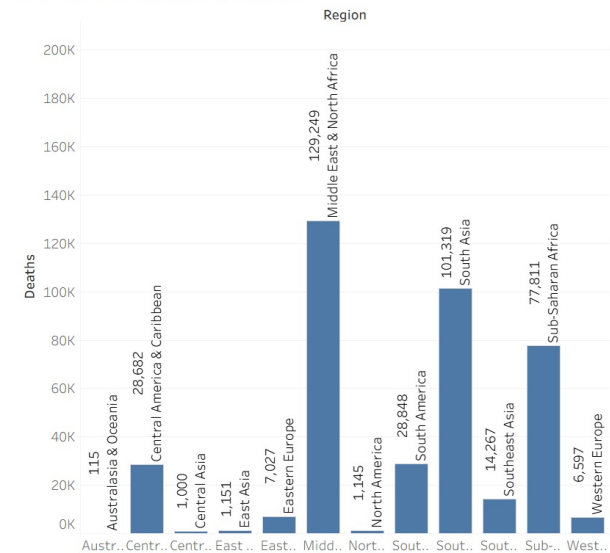


Figure 7: Number of deaths by region

Middle East & North Africa lead with 129,249 deaths, followed closely by South Asia at 101,319. Sub-Saharan Africa records 77,811. Central America & Caribbean (28,682) and South America (28,848) have comparable death tolls. Western Europe, despite its media visibility, has one of the lowest death counts (6,597), reflecting attacks that are frequent but often lower-casualty. North America (1,145) is strikingly low despite the outsized global attention paid to events like 9/11.

(3) Visualizing trends over time

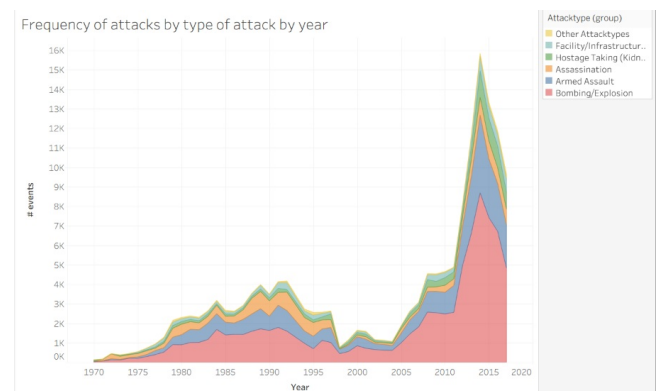


Figure 8: Trend of attack types by year

Bombing/Explosion has consistently been the most prevalent attack method throughout the entire dataset, followed by Armed Assault in second place and Assassination in third. Less frequent attack

types were grouped into an "Other" category to reduce visual noise. Notably, the distribution across attack types has remained relatively stable over time, with no dramatic structural shift in the methods used, the main change has been in the volume of attacks, not their nature.

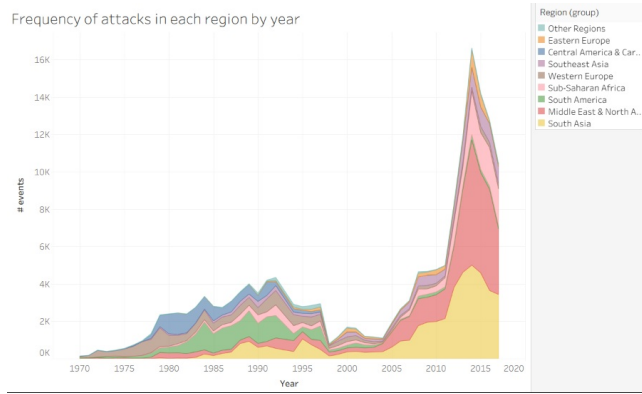


Figure 9: Trend of regions attacked by year

The geographic distribution of terrorism has shifted markedly over the decades. In the early period (1970–1980), Western Europe was the most affected region, but recorded almost no events after approximately 1997. Between roughly 1988 and 1992, Central America & Caribbean and South America saw elevated attack counts, followed by a sharp decline. Meanwhile, South Asia and the Middle East & North Africa had minimal presence in the early dataset but have dominated since approximately 2005, now accounting for the vast majority of recorded events. The global center of terrorism has effectively shifted from the West to the Middle East and South Asia.

4 Dataset Country-Year: V-Dem Core version 16

The V-Dem dataset consists of global political and institutional indicators over time (country-year), covering the evolution of political regimes and the quality of democracy; Each row corresponds to a country-year observation. The most significant variables (which will be selected, with the rest removed) are 'country' - country name, 'year' - year of observation, 'v2x_regime' - type of regime (where 0 corresponds to a closed autocracy, 1 to an electoral autocracy, 2 to an electoral democracy, and 3 to a liberal democracy), 'v2x_polyarchy' - democracy index, 'v2x_civlib' - civil liberties index, 'v2x_rule' - rule of law index, 'v2xel_frefair' - free and fair elections index, 'v2x_clphy' - government violence index, 'v2x_clpol' - political repression index, 'v2x_freexp_altnf' - freedom of expression and access to information index, 'v2x_corr' - corruption level index. The indices range from 0 to 1. The spatial data consist of the variable 'country' (country name), with global coverage at the national level (it does not include cities or regions within the country). The temporal data consist of the 'year' variable (year of observation), with annual frequency and a time range from 1789 to 2025 (after data cleaning, the range becomes 1970–2017, except for 1993), which allows for the analysis of the evolution of political regimes over time. The original dataset contains 28,093 rows, with over 400 variables and 28,902 samples (country-year

observations)—equivalent to the number of rows minus one row corresponding to the header—and is approximately 12 MB in size in ZIP format. It is available for download on the official V-Dem website at [6]. After data cleaning, which is explained below, the dataset is reduced to 7,922 rows, with 11 variables, 7,921 samples, and 375 KB in the xlsx version of the file.

4.1 Index table about the most significant variables

Table 2: V-Dem Dataset Variables Description

Variable	New Name	Description	Type	Example
country	—	Country name	string	Portugal
year	—	Year	int	2005
v2x_regime	regime	Type of government (democracy/autocracy)	cat.	2
v2x_polyarchy	democracylevel	Democracy Index	float	0.78
v2x_civlib	civilliberty	Civil Liberties Index	float	0.65
v2x_rule	lawstate	Rule of Law Index	float	0.70
v2xel_frefair	freefairelections	Index of Free and Fair Elections	float	0.80
v2x_clphy	govviolence	Government Violence Index	float	0.20
v2x_clpol	politicrepression	Political Repression Index	float	0.35
v2x_freexp_altnf	libertyexpression	Freedom of Expression Index	float	0.75
v2x_corr	corruption	Corruption Index	float	0.40

These variables from the V-Dem dataset are useful as a complement to the main dataset, as they allow us to identify patterns in terrorist attacks based on a country's regime, its policies, and its level of freedom. The 'democracylevel' variable is the primary measure of democracy. The 'regime' variable, in turn, allows for comparisons across groups. The variables 'civilliberty', 'lawstate', and 'freefairelections' measure institutional quality. 'govviolence' and 'politicrepression' measure repression (an important factor in analysing terrorism). 'libertyexpression' and 'corruption' account for reporting biases and weak governance.

4.2 EDA details and data cleaning

The primary objective of the data cleaning process for the V-Dem dataset, using Python, was to reduce the complexity of the data and select only the variables relevant to the analysis of political regimes and their characteristics. Initially, variable selection was performed, eliminating the vast majority of the original columns and retaining only a limited set of indicators considered essential, which are the variables presented in Table 2. Next, the variables were renamed to make the names more intuitive and facilitate their use in subsequent stages of the analysis. The updated names are also listed in Table 2. A temporal filter was also applied, restricting the dataset to the period between 1970 and 2017, excluding the year 1993, to ensure compatibility with the main dataset (GTD). Regarding missing values, an exploratory analysis was conducted, calculating the number of missing values per variable using aggregation functions. This allowed us to identify the missing data and its distribution within the dataset. No transformations or removals were applied at this stage, avoiding premature loss of information and allowing for greater flexibility in subsequent stages. Additionally, data types were verified and adjusted to ensure that variables such as year are treated as integer values and indices as continuous numerical variables.

To enable integration between the V-Dem dataset and the GTD, it was necessary to standardize country names. Inconsistencies

were identified between the datasets, such as differences in nomenclature, the inclusion of historical or defunct countries, and the inclusion of territories that do not correspond to sovereign states. To ensure a correct merge, transformations were applied to country names, changing old designations to current equivalents and removing observations without a valid match in the V-Dem dataset. This prevented the introduction of missing values resulting from inconsistencies between datasets.

(1) **How are political regimes distributed across country-year observations?**

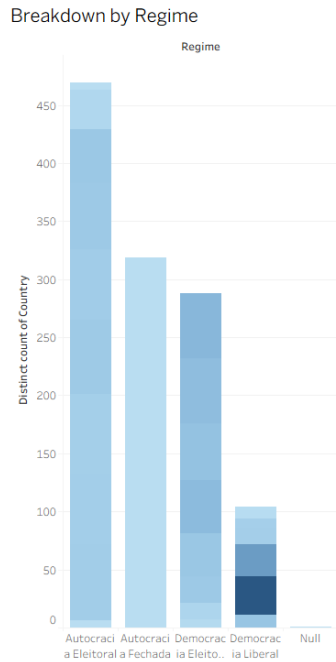


Figure 10: Breakdown by regime

Electoral autocracies and closed autocracies account for a substantial share of country-year observations, with closed autocracies appearing as the most represented category. Electoral autocracies also show a high number of observations, followed by electoral democracies. Liberal democracies are the least represented group. This distribution reflects the global political landscape between 1970 and 2017, where non-democratic regimes were relatively more prevalent, particularly in earlier decades.

(2) **How are democracy and other political indices distributed across all observations?**

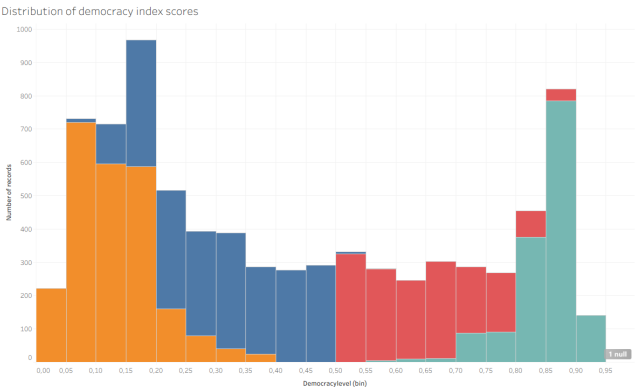


Figure 11: Distribution of democracy index scores

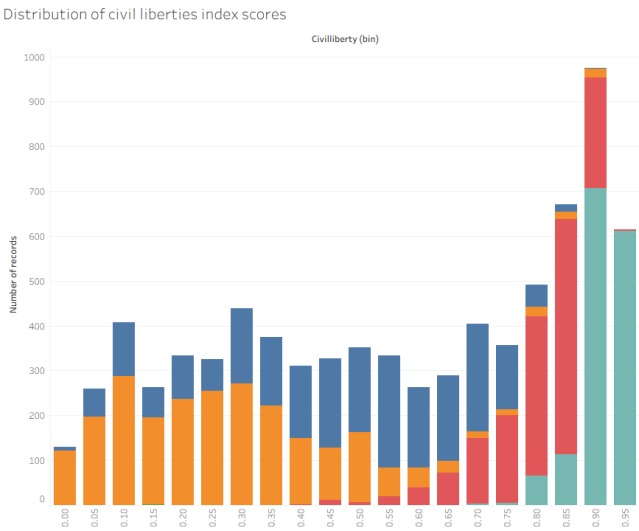


Figure 12: Distribution of civil liberties index scores

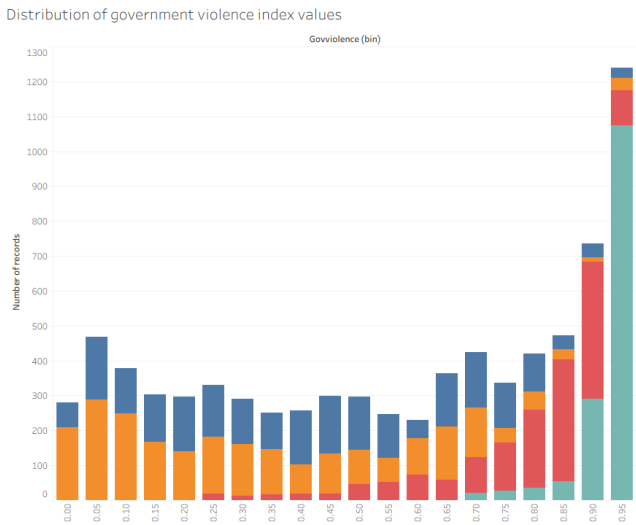


Figure 13: Distribution of government violence index values

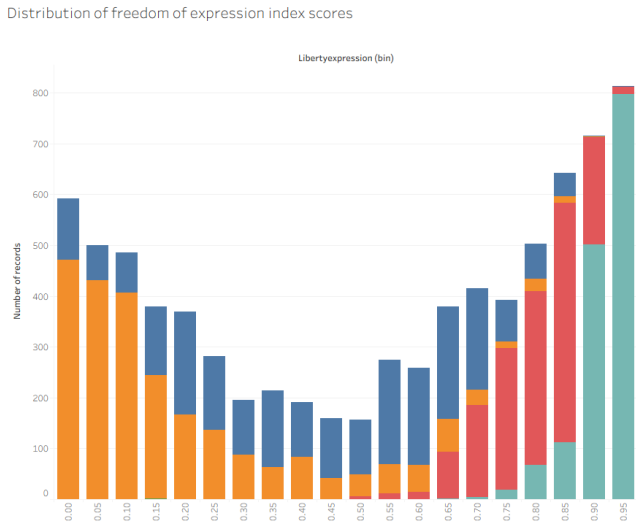


Figure 15: Distribution of freedom of expression index scores

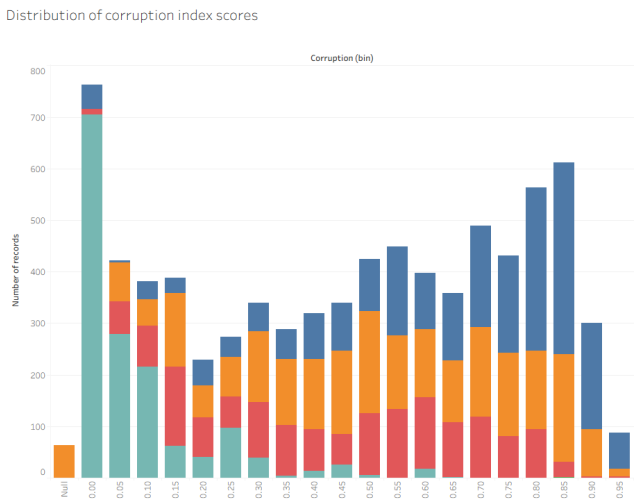


Figure 14: Distribution of corruption index scores

The democracy index shows a bimodal distribution, with concentrations at low values (0.05–0.20) dominated by autocracies and at high values (0.85–0.95) dominated by liberal democracies. Civil liberties follow a similar pattern but with a stronger peak at high values. Government violence is heavily skewed toward high values (0.90–0.95), indicating that most country-year observations record low physical violence by governments. Corruption shows a more uniform distribution across the full range, suggesting it is not strongly tied to regime type alone. Freedom of expression is skewed toward lower values, reflecting limited press freedom across much of the dataset’s time span.

(3) How do political indices vary within and between regime types?

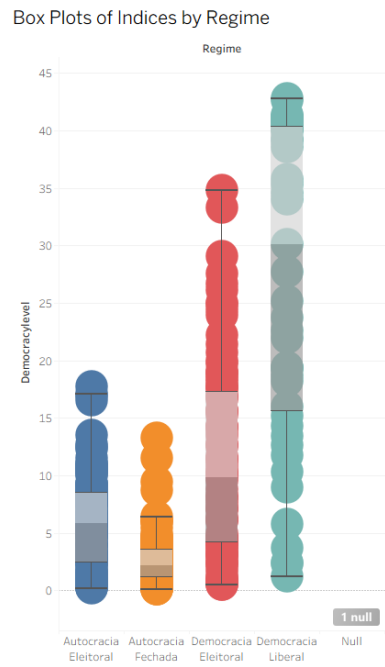


Figure 16: Box plots of indices by regime (democracy level)

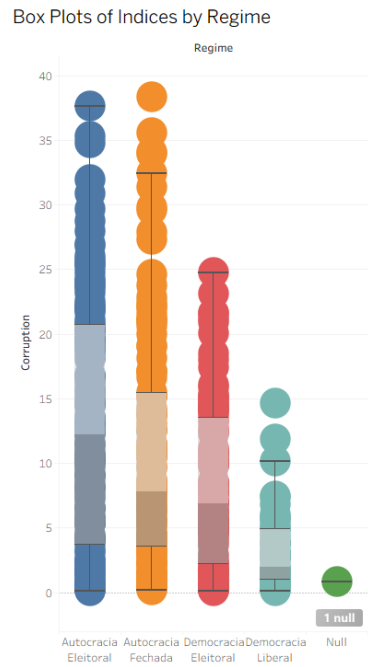


Figure 18: Box plots of indices by regime (corruption)

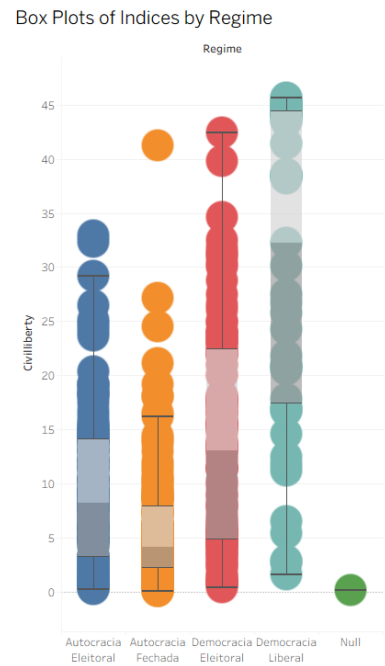


Figure 17: Box plots of indices by regime (civil liberty)

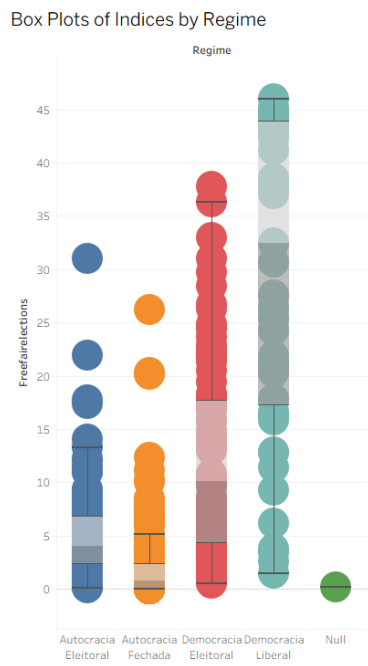


Figure 19: Box plots of indices by regime (free fair elections)

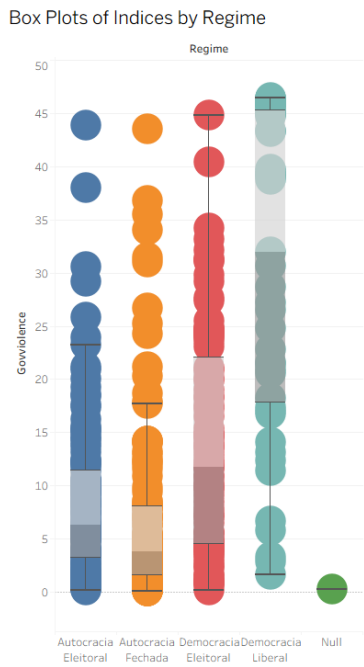


Figure 20: Box plots of indices by regime (govern violence)

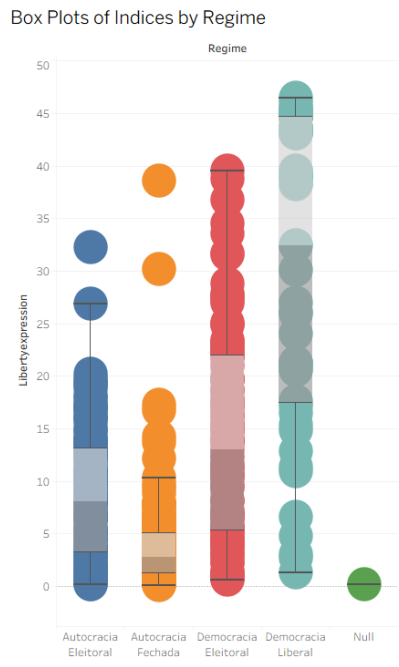


Figure 22: Box plots of indices by regime (liberty of expression)

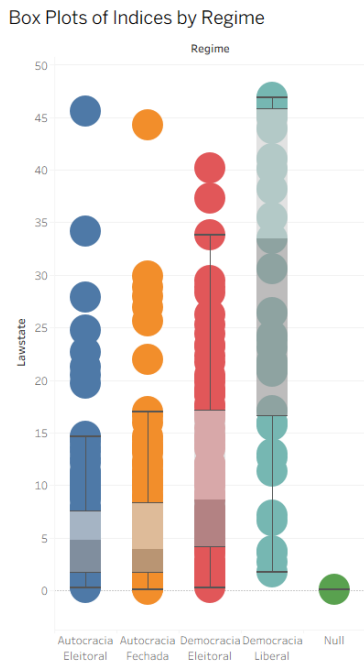


Figure 21: Box plots of indices by regime (law state)

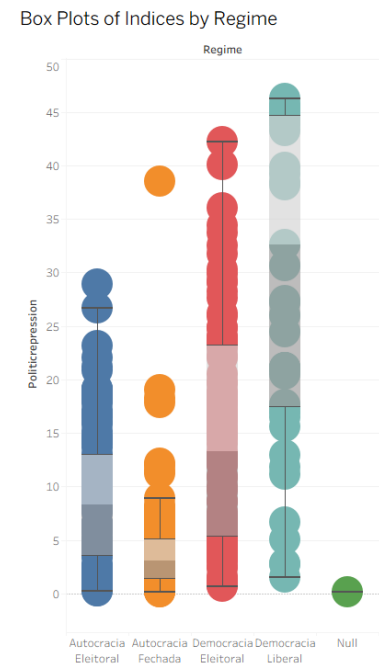


Figure 23: Box plots of indices by regime (politic repression)

Box plots confirm clear separation between regime types across all indices. Liberal democracies consistently score highest on democracy level and civil liberties, and lowest on government violence and

political repression. Closed autocracies show the opposite pattern. Electoral autocracies and electoral democracies occupy intermediate positions, with considerable overlap and internal variance, suggesting these categories are more heterogeneous. Corruption is the least discriminating variable, with notable spread across all regime types.

(4) Are there missing values that could affect the analysis?

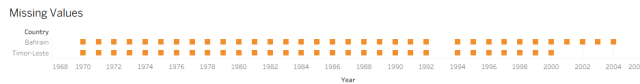


Figure 24: Missing values

Missing values are limited and concentrated in two variables: 'corruption' (64 missing values) and 'lawstate' (6 missing values). These are largely attributable to two countries, Bahrain and Timor-Leste, which lack data for certain years. Given the small number of missing values relative to the total dataset size, their impact on the analysis is expected to be minimal.

(5) Are there missing values that could affect the analysis?

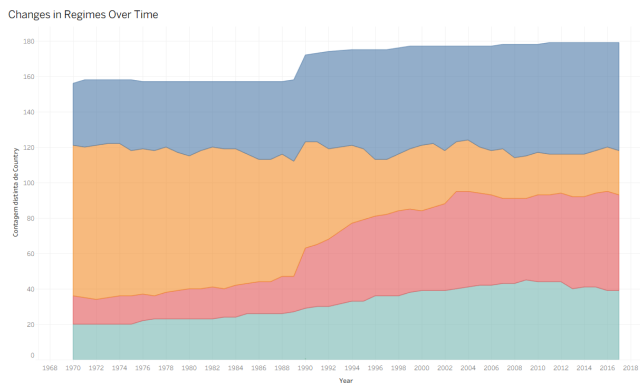


Figure 25: Changes in regimes over time

The area chart reveals a significant shift in global regime composition over the period. In 1970, closed and electoral autocracies dominated. From the late 1980s onward, a notable expansion of electoral and liberal democracies is visible, consistent with the third wave of democratisation. By the 2000s, the total number of democracies seems to surpass autocracies. However, from around 2012 onward, the share of liberal democracies begins to stagnate or slightly decline, hinting at early signs of democratic backsliding.

(6) Which countries and regions are predominantly autocratic or democratic?

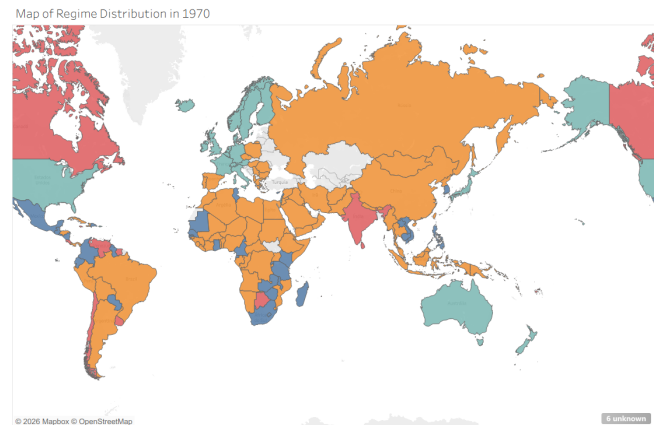


Figure 26: Map of regime distribution in 1970

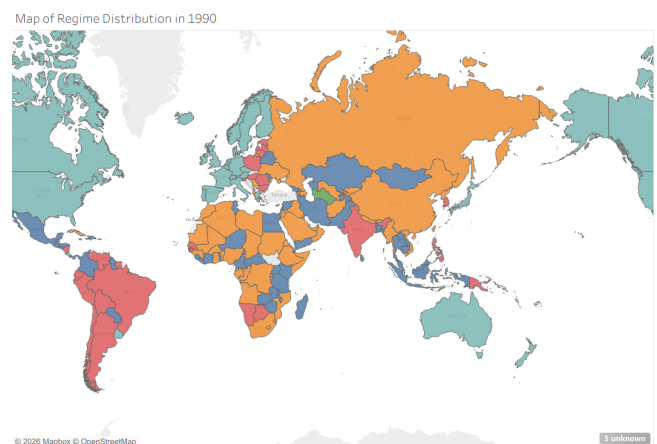


Figure 27: Map of regime distribution in 1990

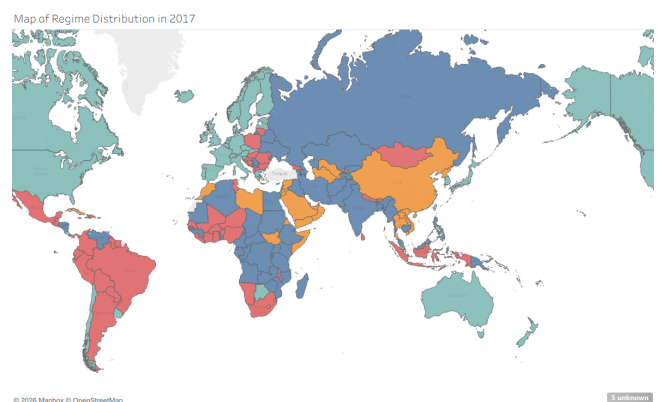


Figure 28: Map of regime distribution in 2017

The maps for 1970, 1990, and 2017 illustrate the geographic transformation of political regimes over time. In 1970, autocracies dominated across Africa, Asia, and Latin America, with democracies

largely confined to Western Europe, North America, and Oceania. By 1990, the collapse of communist regimes in Eastern Europe began reshaping the map. By 2017, electoral democracies had expanded significantly across Latin America, Sub-Saharan Africa, and parts of Asia, though autocracies remained concentrated in the Middle East, Central Asia, and parts of Africa.

(7) Do regime types show consistently different levels of civil liberties, repression, and corruption?



Figure 29: Average indices by regime

The average indices by regime chart confirms that regime type is a strong predictor of institutional quality. Civil liberties, freedom of expression, and rule of law increase consistently from closed autocracies to liberal democracies. Conversely, government violence and political repression decrease as regimes become more democratic. Corruption is the exception, while liberal democracies score lowest, the difference across groups is smaller and less consistent, suggesting corruption is influenced by additional factors beyond regime type alone.

(8) Is there a relationship between democracy level and corruption across countries?

Democracy vs Corruption Index

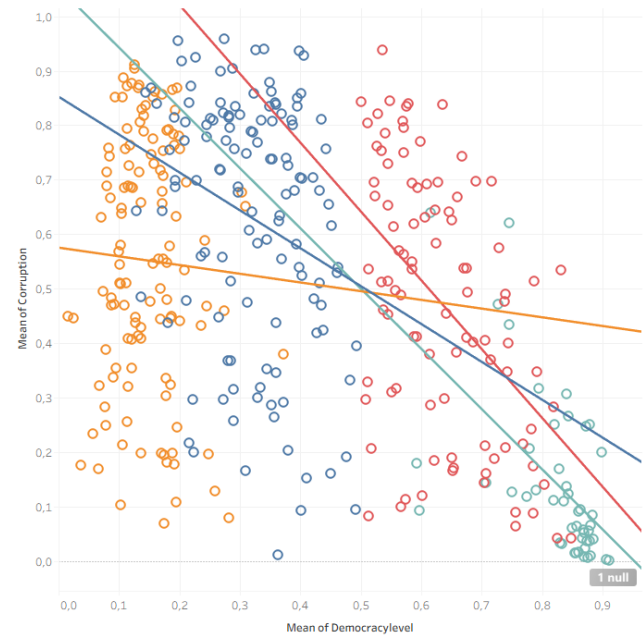


Figure 30: Democracy vs Corruption index

Government Violence vs Political Repression Index

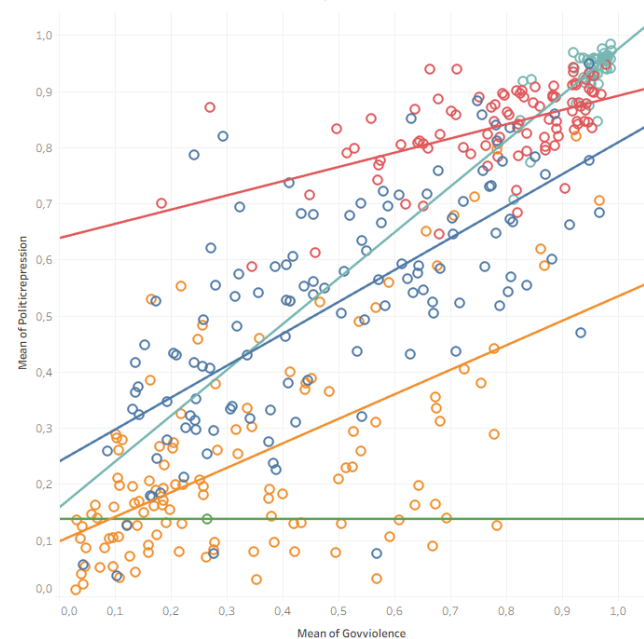


Figure 31: Government violence vs Political repression index

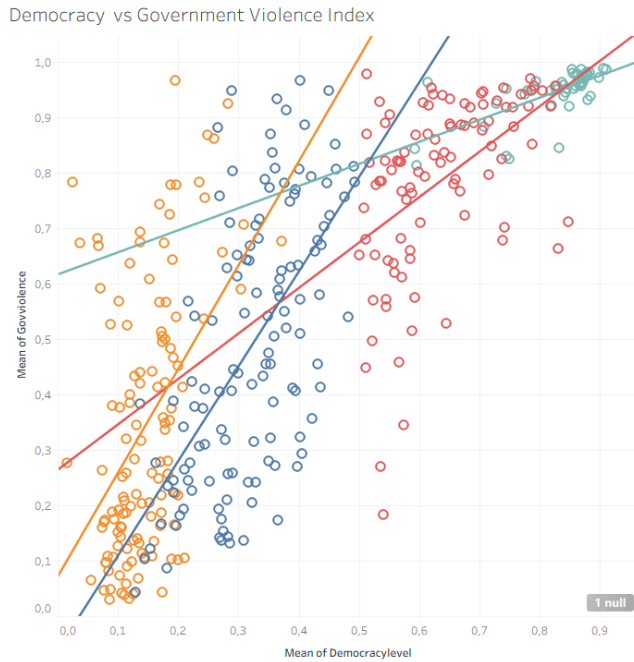


Figure 32: Democracy vs Government index

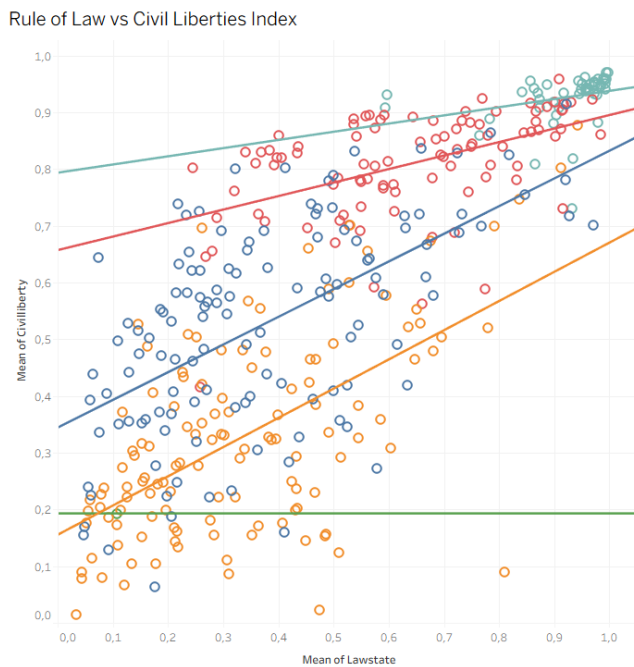


Figure 33: Rule of law vs Civil liberties index

The scatter plots reveal clear negative relationships between democracy level and corruption, and between democracy level and government violence (countries with higher democracy scores tend

to have lower corruption and less state violence). Government violence and political repression are strongly positively correlated, suggesting they co-occur as instruments of authoritarian control. Rule of law and civil liberties show a strong positive correlation, indicating these institutional dimensions reinforce each other. These relationships validate the use of V-Dem indices as complementary variables to contextualize terrorism patterns in the main dataset.

5 State of the art

Before conducting any data analysis or developing visualisations, it is essential to understand the scientific context of the phenomenon under study and to examine existing work related to the topic. The purpose of the state of the art section is precisely to present the current knowledge surrounding terrorism, identify approaches used by other authors, and understand how different analytical and visualisation techniques have been applied to the study of this phenomenon.

This section is important for several reasons. First, it allows the research questions proposed in this work to be contextualized within the existing literature. Second, it helps identify methodologies, variables, and patterns that are frequently explored in previous studies. Finally, it provides inspiration and justification for the visualisation and analytical techniques adopted throughout the project.

Considering the objectives of this work, the state of the art is divided into two main components:

- Scientific contextualization of terrorism and its relationship with political factors;
- Existing studies and visualisations related to terrorism and associated data analysis.

5.1 Terrorism and Political Context

Terrorism can be understood as the deliberate and planned use of violence, or the threat thereof, by individuals, groups or states, with the aim of intimidating populations and achieving political ends [11]. It constitutes a form of action that systematically resorts to force, and is distinguished from an ideology in itself, it is rather a method, with its own tactics and strategies oriented towards conflict [11]. Its targets tend to be random or indiscriminate, frequently resulting in the victimisation of innocent civilians, making it one of the greatest threats to fundamental human rights, namely the right to life and the right to live free from fear [3].

The absence of a universal and legally binding definition remains one of the central challenges in the study of terrorism [18]. This difficulty stems from multiple converging factors. There is, first and foremost, a deep contradiction in the perceptions of the phenomenon: politicians, legal scholars and human rights activists frequently diverge on what constitutes a terrorist act and who should be considered a terrorist [18]. Furthermore, terrorism is always underpinned by an ideological motivation, and since the objectives, targets and methods of different organisations vary considerably, any broad definition risks encompassing activities that ought to be addressed under international humanitarian law [18]. The high subjectivity of the concept also leads each state to define terrorism in accordance with its own national interests [18]. In the absence of definitional consensus, gaps and inconsistencies in the application of the law inevitably arise [18][4].

When terrorism is analysed as rational and intentional behaviour, it becomes possible to understand the ideological justifications attributed to it by the groups that practice it [17]. Ideologically motivated terrorist organisations resort to violence to achieve concrete political outcomes, convey messages, punish societies perceived as adversarial, or obtain concessions through the threat of harm [17]. Within this framework, the selection of specific targets is designed to reinforce collective fear and demonstrate the state’s inability to protect its citizens [17]. Academic research has identified a wide range of explanatory factors for terrorism, spanning socioeconomic conditions, government behaviour, modernisation, religion, ethnic relations, historical injustices, conflicts and personality traits [15]. Two dominant interpretive strands have emerged: one that conceives of terrorism primarily as a strategic and rational action by groups pursuing political objectives, and another that emphasises individual psychological dynamics, such as identity construction, the desire for self-reinvention and the sense of belonging to a group [15]. More recent authors argue that these two dimensions are complementary, with terrorism being simultaneously a product of individual motivations and strategic organisational considerations [15].

The relationship between terrorism and political regime type has been the subject of extensive study, with frequently contradictory results. Several studies identify a positive association between democracy and terrorism, grounded in four main explanations [19]. The structural explanation rests on the idea that the institutional design of democracies itself offers opportunities that facilitate the execution of terrorist acts [19]. The strategic explanation assumes that terrorists prefer to target democracies because these are more susceptible to conceding to their demands, although this thesis lacks robust empirical support, as democracies tend to harden their responses following initial concessions [19]. The organisational explanation suggests that democracies, being more permeable to political mobilisation, provide fertile ground for the emergence of terrorist organisations, though empirical tests of this hypothesis yield mixed results [19]. Finally, the political explanation links the intensity of terrorism to the specific policies adopted by governments, such as military occupation, which may intensify the mobilisation of the population against the regime [19]. On the other hand, other studies find a negative relationship between democracy and terrorism, arguing that democratic institutions ensure the representation of minorities and limit the resort to violence as a form of political expression [19].

The historical record reveals, however, that towards the end of the twentieth century terrorism occurred more frequently in democratic countries than in authoritarian regimes [5]. Evidence also suggests that intermediate regimes, referred to as "anocracies" or partial democracies undergoing transition, tend to register the highest levels of terrorist activity [5]. Some studies have also identified differences according to governmental orientation: left-wing governments appear to be more frequently targeted by transnational attacks, while right-wing regimes may be more vulnerable to domestic terrorism [5]. Totalitarian regimes, by contrast, possess far more effective instruments of control, from harsher interrogation methods to the capacity to pursue dissidents abroad, which tends to inhibit the emergence of terrorist groups within their borders

[21]. Political systems in transition present heightened vulnerability: as the coercive power of the old regime’s security forces wanes, the new freedoms open space for violent dissidents before new institutions are fully consolidated [21]. It is also worth noting that historical democratisation has not necessarily translated into a reduction in terrorism, on the contrary, evidence suggests that periods of democratic transition may be associated with a temporary increase in terrorist activity [21].

5.2 Previous studies and visualisations

Data visualisation plays a fundamental role in the interpretation and communication of complex information, particularly when dealing with large-scale datasets and temporal patterns. Effective visual representations allow researchers and analysts to identify trends, correlations, anomalies, and behavioural changes that might otherwise remain difficult to perceive through raw numerical data alone. In the context of terrorist organisations and their life cycles, visualisation methods have also been explored in previous studies. Figures 34 and 35 present the evolution of major terrorist groups over time, considering indicators such as the number of attacks performed and the number of fatalities caused [8]. These representations make it possible to observe patterns of emergence, expansion, operational peaks, fragmentation, and decline throughout different historical periods.

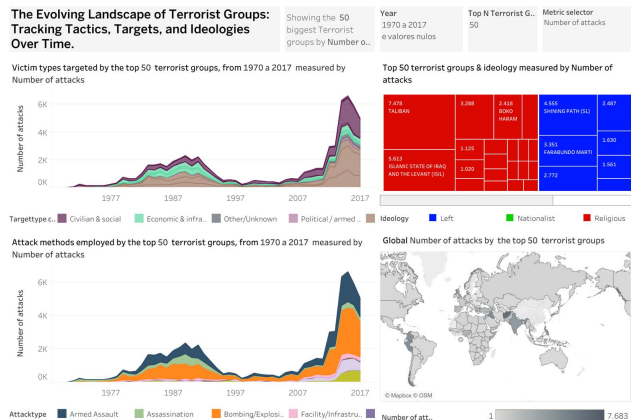


Figure 34: Evolution of principal terrorism groups

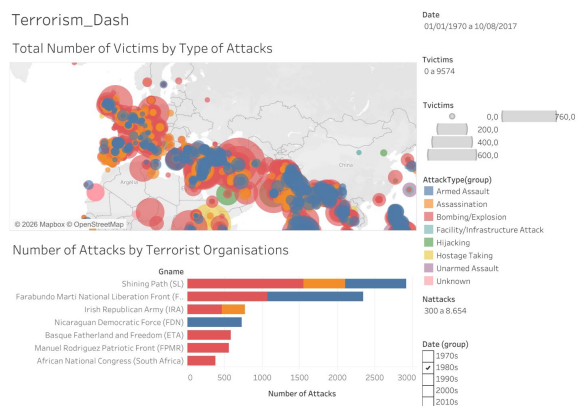


Figure 35: Number of attacks over time by 10 principle terrorist groups

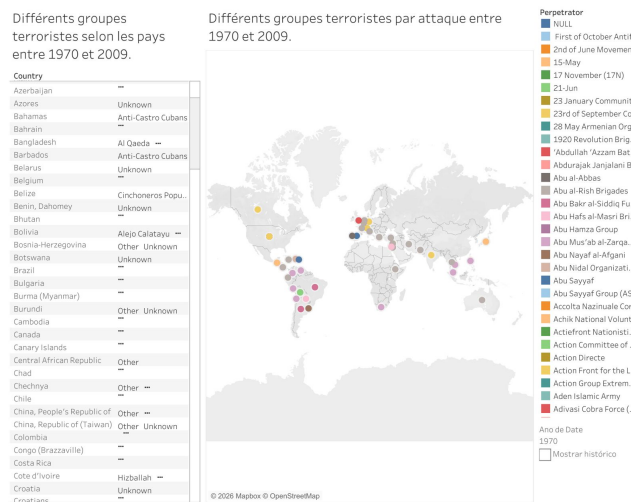


Figure 36: Number of attacks over time by 10 principle terrorist groups

The two charts from Figure 34 and 35 reveal a stark transformation in the global terrorist landscape from the 1990s onward, marked by the rise of a small number of extraordinarily lethal and prolific organisations. While earlier decades were characterized by moderate, relatively dispersed activity from groups such as the IRA, FARC, Shining Path, and the FMLN, the post-2010 period witnesses an unprecedented surge driven almost exclusively by ISIL, Boko Haram, and the Taliban. ISIL alone surpasses 11,000 deaths in a single year by 2014–2015, a figure unmatched by any group in the entire dataset, while Boko Haram peaks at over 6,000 deaths around the same period. In terms of attack volume, ISIL and the Taliban similarly dominate post-2010 activity, each exceeding 1,000 attacks per year at their peak. However, existing approaches focus almost exclusively on the temporal evolution of these organisations in terms of attack volume and lethality, and no visualisations were

found addressing the regions affected by each terrorist group, nor the types of attacks and respective weapons associated with them. Figure 36 illustrates one of the few existing attempts in this direction, mapping the dominant terrorist groups by country attacked between 1970 and 2009, but without any dimension of lethality, weapon type, or temporal filtering by group. Through the Extended Data on Terrorist Groups (EDTG) [16], advance in this direction by analysing how group ideology influences regional location, attack characteristics, and organisational longevity [16], but their analysis remains essentially statistical and tabular, without an integrated visual representation allowing the simultaneous exploration of all these dimensions for a given group. [12] proposed a unified visualisation environment for terrorism incident data combining spatial, temporal, and multivariate patterns [12], but this work was developed with a far more limited version of the database (27,000 incidents up to 2006) and does not support interactive filtering by group, nor does it cross all operational dimensions simultaneously. Understanding where attacks occur, as well as the methods and weapons most frequently employed, can provide a more comprehensive perspective on the strategies, capabilities, and areas of influence of terrorist organisations, aspects that will be explored in greater detail in Chapters 6 and 7. Previous studies and visualisations have already explored attack types, target types, and the main regions affected by terrorist activity, with these dimensions mainly analysed according to the number of events per year, as seen in Figures 37 and 38 [2][24].

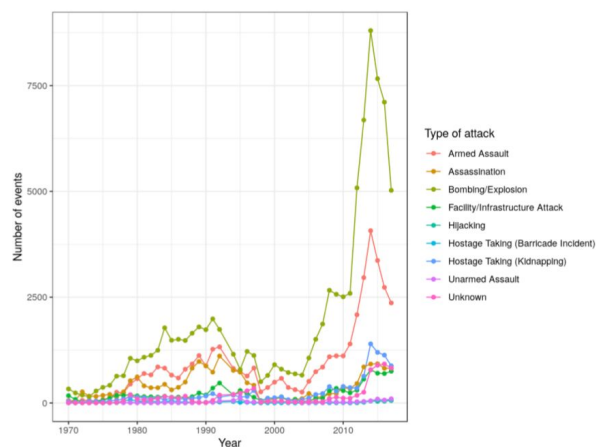


Figure 37: Evolution of attack types over the decades

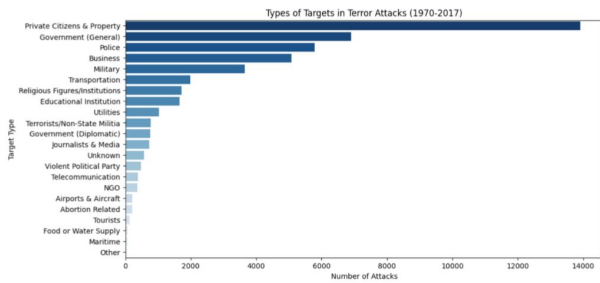


Figure 38: Attack targets

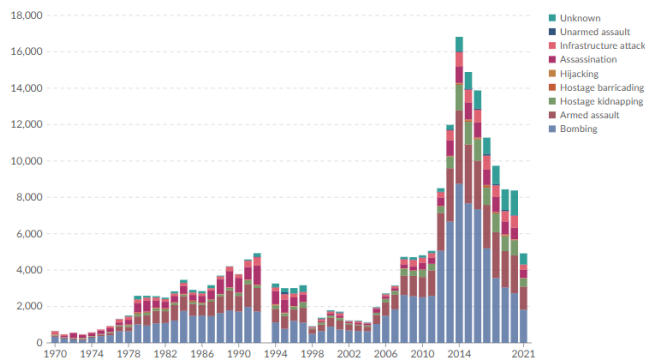


Figure 39: Deaths from terrorist attacks by method of attack, globally, 1970–2020

The data from Figure 37 reveal that bombing and explosion is by far the dominant attack modality across the entire 1970-2017 period, with a dramatic surge peaking around 2013-2014 that dwarfs all other attack types, consistent with the ISIS-driven escalation identified in the preceding analysis, before declining sharply thereafter. Armed assaults follow as the second most frequent method, also rising significantly in the post-2010 period, while all other modalities (assassinations, facility attacks, hijackings, hostage-taking) remain comparatively marginal and relatively stable over time. This temporal dominance of bombing is further corroborated by Figure 39, which presents the global distribution of terrorism deaths by method of attack, confirming that explosive-based attacks account for the vast majority of fatalities recorded in the GTD across the full dataset period [14]. Regarding targeting, private citizens and property constitute the most frequent target category by a considerable margin (nearly 14,000 attacks), in Figure 38, followed by general government institutions, police, and businesses, a pattern that underscores how modern terrorism has shifted away from purely symbolic or state-directed violence toward maximizing civilian casualties and societal disruption. However, the literature addresses attack type and target type largely in isolation, existing visualisations that analyse the frequency of events per attack type over the years, or the distribution of the most common target categories, but do not cross these two dimensions with associated lethality. This absence is significant, knowing that bombing is the most frequent attack type is not equivalent to knowing which combinations of

attack and target produce the highest number of deaths per incident. [22] demonstrate that the distribution of injuries and deaths varies considerably by weapon type and world region [22], reinforcing the need for an analysis that jointly combines attack type, target type, and lethal impact. No visualisation found in the literature presents this triple combination in an exploratory and interactive form. For this reason, conducting an analysis that combines both the frequency of events and their lethality is considered relevant for achieving a more comprehensive and robust understanding of terrorist activity patterns [2][24], that is addressed in Dashboard 3 of this work.

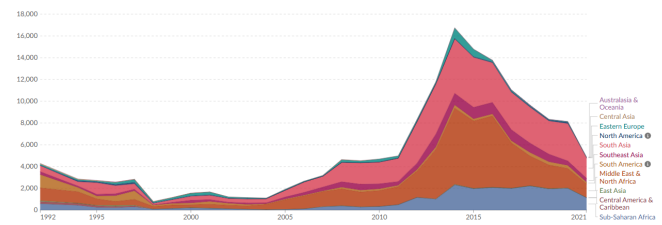


Figure 40: Terrorist attacks by region over time

Figure 40 illustrates the temporal evolution of terrorist attacks by region between 1992 and 2021, revealing a sharp global peak around 2014-2015, driven predominantly by activity in the Middle East & North Africa and Sub-Saharan Africa, followed by a gradual decline in subsequent years.

The relationship between the type of political regime and terrorist activity has been the subject of extensive study in the academic literature, but almost exclusively through statistical methods, without the use of interactive visualisations that allow these relationships to be explored intuitively. [20] demonstrated that restrictions on the executive branch increase the frequency of terrorist attacks, while political participation reduces it, thus identifying opposing effects within democracy itself [20]. [10] expanded on this analysis by empirically demonstrating that the relationship between regime type and terrorism follows an inverted U-curve, with intermediate regimes (called anocracies or partial democracies) registering the highest levels of terrorist activity [10]. [13] confirm this ambivalence, concluding that different dimensions of democracy both stimulate and suppress the terrorist impulse, without the literature having reached a definitive consensus [13]. Despite the robustness of these empirical results, none of these studies produces visualisations that allow the user to interactively explore the distribution of attacks, deaths, or success rates by regime type over time. The integration of V-Dem indices with GTD data into an exploratory visual platform is, therefore, an original contribution of this work, implemented in Dashboards 5, 6, 7 and 8.



Figure 41: Map of success rates

Regarding the study of the success of terrorist attacks, visualisations addressing this topic were found in the literature, particularly through geographical representations such as the one illustrated in Figure 41, which allow the identification of regional patterns and variations in success rates across different geographical areas [23]. The choropleth map shows that terrorist attacks succeed at alarmingly high rates virtually everywhere in the world, with success rates clustering near 0.8 to 1.0 across the vast majority of states regardless of regime type or geographic region. This near-universal pattern suggests that once an attack is initiated, prevention at the point of execution remains extremely difficult across all political contexts, and underscores that counter-terrorism effectiveness must be evaluated not only at the point of attack, but across the full causal chain from radicalization to execution. However, these geographical representations do not relate attack success to institutional variables such as democracy level, rule of law, political repression, or freedom of expression, all available in the V-Dem dataset. [7] suggest that coercive government responses tend to contribute to the decline of terrorist campaigns [7], but this hypothesis has never been systematically visualized by crossing continuous institutional indicators with per-attack success data. This gap is addressed in Dashboard 8, which present scatter plots, heatmaps, and maps that allow users to interactively explore the relationship between institutional quality and attack success rates.

In summary, the existing literature on terrorism data visualisation has made significant progress in representing temporal trends, geographic distributions, and the profiles of major terrorist groups. However, three key analytical gaps persist: (1) the absence of integrated multi-dimensional visualisations of terrorist group lifecycles combining region, tactic, weapon type, and lethality; (2) the lack of interactive tools that jointly analyse attack type, target type, and lethality per incident; and (3) the near-total absence of visual representations that connect political regime indicators, particularly those from V-Dem, with terrorism frequency, lethality, and attack success rates. The present work directly addresses these three gaps through a set of eight interactive dashboards built in Tableau.

6 Data visualisation

In order to answer the three Research Questions and each sub-question, some graphs were created to support the final dashboards. In this section, all the graphs that helped to analyse the data will be presented.

6.1 What is the lifecycle of a terrorist group?

(1) When do terrorist groups begin and cease their activity?

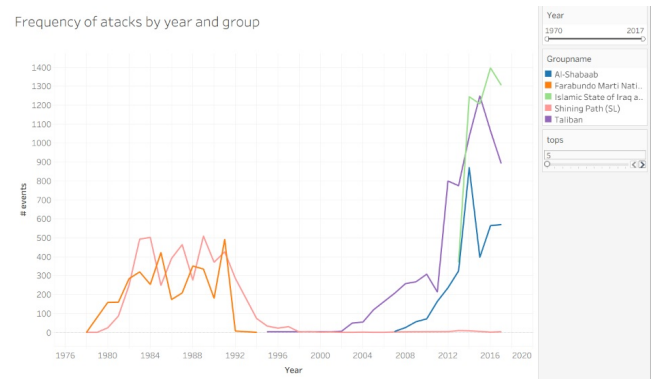


Figure 42: Top 5 groups with more activity through time

This view applies two filters: only groups with more than 10 total attacks are included (to exclude marginal actors), and the display is further limited to the top 5 groups by attack count in the image. This allows for a clear reading of each group's activity window: when they emerged, when they peaked and in some cases when they ceased operations. Shining Path (SL) was most active between approximately 1982 and 1991, with a notable surge and subsequent decline during those years. The group remains nominally active today but with very limited activity. The Islamic State of Iraq and the Levant (ISIL) began its recorded activity around 2012 and experienced an exponential rise almost immediately, becoming the most active group in the dataset's recent years and peaking at approximately 1400 attacks in 2017.

(2) What types of attacks are most commonly associated with each group?

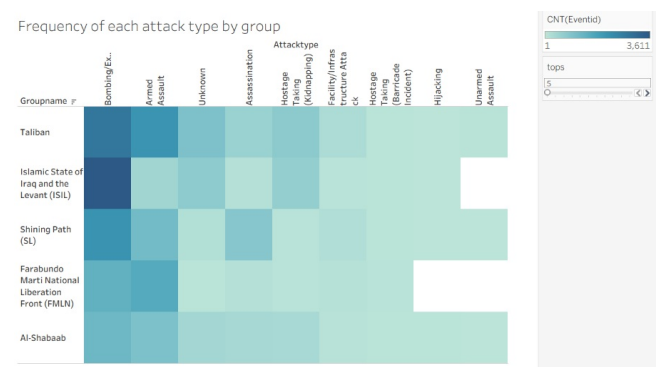


Figure 43: Types of attacks associated with each group - Top-5 groups

Bombing/Explosion is the dominant method across most top-5 groups, with ISIL being its most prolific user. The exception is

FMLN, which relies more heavily on Armed Assault. Notably, neither ISIL nor FMLN have any recorded Unarmed Assault attacks in the dataset.

(3) Which regions are most frequently targeted?

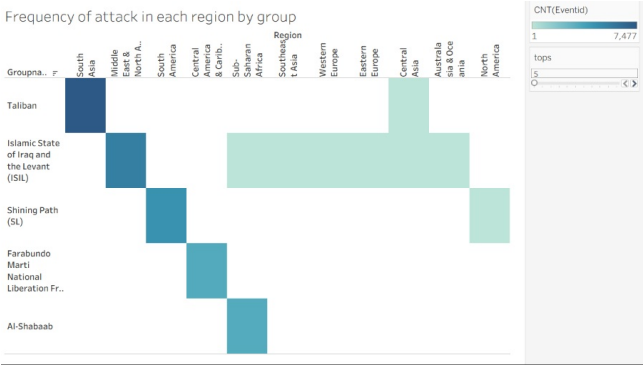


Figure 44: Regions with more attacks - Top-5 groups

Regions and groups are ordered from most to least frequent. The analysis is filtered to the top 5 groups by attack count. Most of these groups operate predominantly within one or two regions, with a clear geographic anchor. The exception is ISIL, which carried out attacks across nearly all regions, only South Asia, South America, Central America & Caribbean, and North America show no recorded ISIL attacks. The regional breakdown for each group’s primary area of activity is: Taliban -> South Asia; ISIL -> Middle East & North Africa; Shining Path -> South America; Farabundo Marti National Liberation Front (FMLN) -> Central America & Caribbean; Al-Shabaab -> Sub-Saharan Africa.

(4) What are the typical levels of lethality (e.g., number of casualties)?

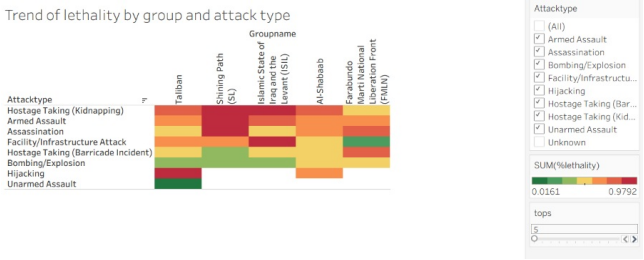


Figure 45: Brutality of each group by attack type

This heatmap reveals the average deadliness of each attack type per group. Broadly, ISIL and Shining Path record the highest lethality in Hostage Taking (Kidnapping) events, these are the scenarios in which the most deaths occur per attack. Taliban stands out as the only group with recorded lethality across every attack type. Most other groups have no recorded Unarmed Assault events, and where Unarmed Assault is present (as in Taliban), it registers the lowest lethality score. Hijacking appears in only two groups’ profiles, but

within Taliban’s data it is actually the most lethal attack method, a notable outlier.



Figure 46: Brutality of each group by weapon type

The handgun emerges as the highest-lethality weapon subtypes relative to their usage, a counter-intuitive finding given that conventional weapons dominate in frequency. Firearms are the weapon types mostly shared across all five top groups, representing the common operational toolkit. Taliban shows the widest variety of weapon usage among the top groups, while FMLN is the most restricted in its arsenal. Most weapon subtypes are not highly lethal in practice, the categories with greater lethality relative to their use are specific firearms, melee weapons, and incendiary devices.

(5) What types of weapons are used?

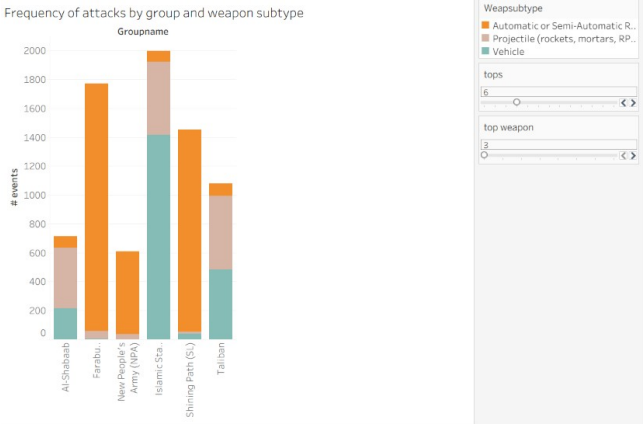


Figure 47: Frequency of attacks by group and weapon subtype

Overall, the top 3 weapons used are Automatic or Semi-Automatic Rifle, Vehicle and Projectile, from Firearms and Explosive categories. FMLN and SL attack mostly with Automatic or Semi-Automatic Rifles, while the other ones tend to use more the other two and less Rifle.

6.2 How do attack type, target type, and region jointly influence the lethality and success of terrorist attacks over time?

(1) Which combinations of attack and target types are associated with higher lethality?

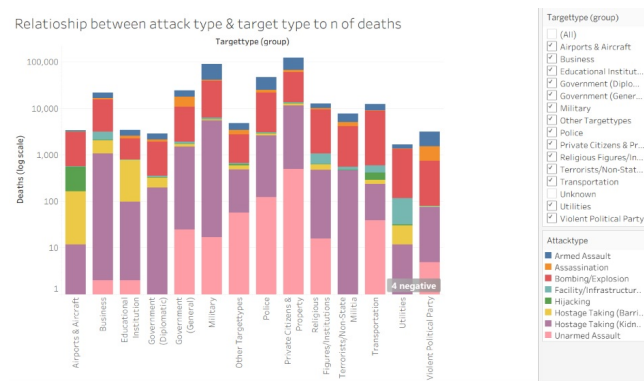


Figure 48: Attack & target types associated with higher lethality

For nearly every target category, Hostage Taking (Kidnapping) is the most frequent attack type, reflecting its use as a tactical tool across diverse contexts. Unarmed Assault appears as a secondary method for targets such as Military, Police and Private Citizens & Property, while Airports & Aircraft, Government (Diplomatic), and Utilities show a higher representation of Bombing/Explosion as the secondary method. As expected, Hijacking is almost exclusively associated with Airports & Aircraft. Assassination shows the highest relative prevalence against Government (General) and Violent Political Party targets, suggesting these attacks are more deliberate and politically motivated in intent.

(2) How does the success rate of attacks vary across regions?

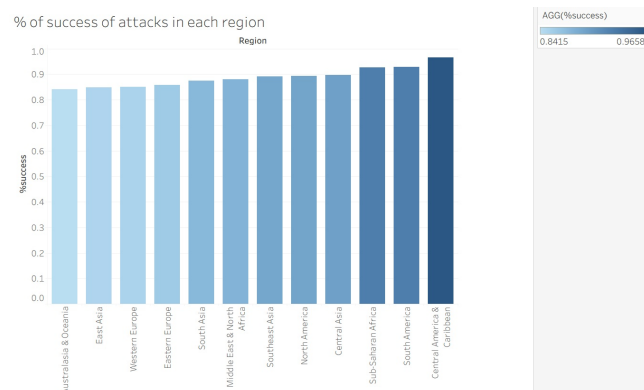


Figure 49: Success of attacks across regions

Australasia & Oceania has the lowest attack success rate in the dataset at approximately 84.2%, indicating that local authorities

are relatively effective at disrupting or containing incidents. At the other end of the spectrum, Central America & Caribbean records a success rate of approximately 96.6%, meaning that in this region, attackers almost always achieve their stated objective.

(3) Are there temporal trends in these relationships?

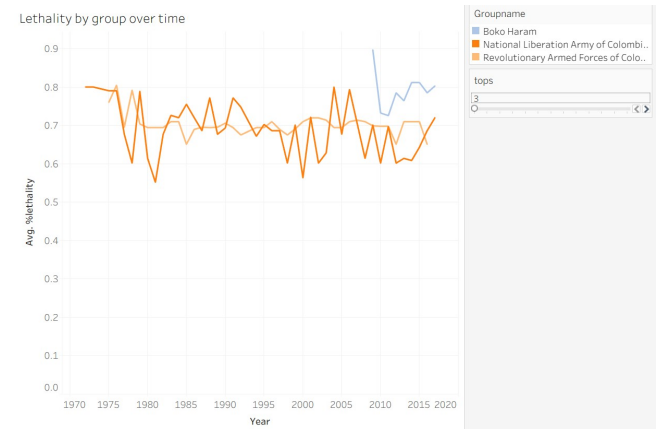


Figure 50: Lethality by group over time

Although it is a recent group, Boko Haram is the most lethal group in these data, with a peak in its first year with almost 90% lethality. The top-3 is also composed of the National Liberation Army of Colombia (ENL) and the Revolutionary Armed Forces of Colombia (FARC), both active since the 1970s.

6.3 How do political regimes shape the frequency, lethality, and success of terrorist attacks?

(1) Are certain types of political regimes associated with higher frequencies of attacks?

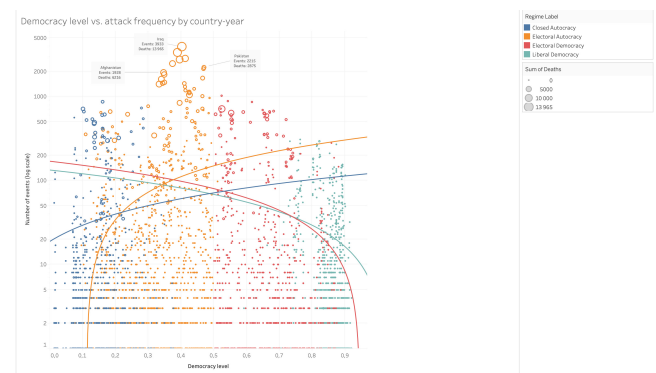


Figure 51: Democracy level vs. attack frequency by country-year

The scatter plot maps the relationship between democracy level (x-axis, ranging from 0 to approximately 0.95) and attack frequency (y-axis, on a logarithmic scale), with each observation representing

a country-year unit. Bubble size encodes total deaths, and colour distinguishes four regime types: Closed Autocracy, Electoral Autocracy, Electoral Democracy, and Liberal Democracy. Trend lines are fitted separately for each regime category, allowing for a direct visual comparison of how political institutions shape terrorist activity. The visualisation reveals a clear and substantively important pattern: attack frequency is not uniformly distributed across regime types, nor does it follow a simple linear relationship with democratisation. Electoral Autocracies consistently occupy the upper range of the distribution, with the most extreme cases: Iraq (3,933 events; 13,965 deaths), Pakistan (2,215 events; 2,875 deaths), and Afghanistan (1,928 events; 6,216 deaths), clustered at the top of the chart and all classified within this regime category. The corresponding trend line remains the highest across nearly the entire democracy spectrum, suggesting that hybrid regimes combining nominal electoral competition with weak institutional constraints are systematically associated with greater terrorist activity. At the opposite extreme, Closed Autocracies display notably low attack frequencies, with their trend line remaining near the bottom of the distribution. This may reflect either genuine suppression of political violence through coercive state capacity, or systematic under-reporting in fully closed political environments. Liberal and Electoral Democracies occupy an intermediate position, with their trend lines declining sharply as democracy levels approach their upper bound, suggesting that consolidated democratic institutions are associated with reduced terrorist activity.

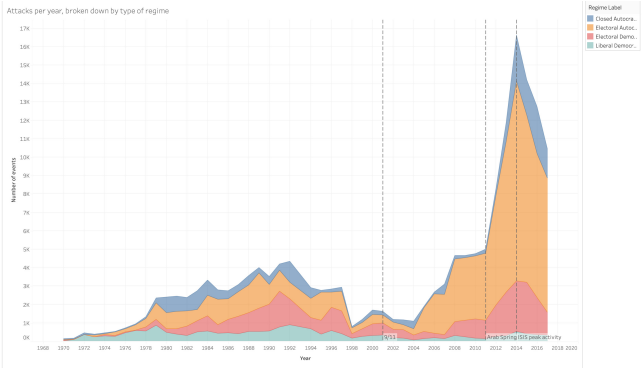


Figure 52: Attacks per year, broken down by type of regime

The stacked area chart displays the total number of terrorist attacks per year from 1970 to approximately 2020, disaggregated by regime type. The chart reveals a striking temporal pattern in which global terrorist activity remained relatively contained until the early 2000s, fluctuating between roughly 2,000 and 4,000 annual events, before entering a sharp and sustained upward trajectory that culminated in an unprecedented spike of over 16,000 attacks around 2014-2015. Crucially, this escalation is driven almost entirely by Electoral Autocracies and, to a lesser but visible extent, Electoral Democracies, while Closed Autocracies and Liberal Democracies contribute a comparatively stable and modest share throughout the entire period.



Figure 53: Democracy level at each country vs. terrorist events by year

This choropleth-bubble displays the geographic distribution of terrorist events across countries, with bubble size encoding the count of events and color shade representing the median regime score on an autocracy-to-democracy continuum. The visualisation thus simultaneously conveys the spatial concentration of political violence and the institutional character of the states in which it occurs. The map immediately makes the geographic and political concentration of terrorism clear. The largest bubbles cluster heavily across the Middle East, South Asia, and Sub-Saharan Africa, with Iraq, Afghanistan and Pakistan standing out as the most affected. These countries are rendered in darker shades, indicating lower democracy scores and greater proximity to the autocratic end of the spectrum, yet they are predominantly hybrid or electoral autocratic regimes rather than fully closed ones. Western Europe, North America, and East Asia, which tend toward higher democracy levels and lighter shading, display only marginal bubble sizes, reflecting comparatively negligible attack frequencies. This spatial pattern reinforces the argument developed in the preceding analyses: terrorist activity is not randomly distributed across the international system, but is instead structurally concentrated in states that combine partial political opening with weak institutional capacity.

(2) Do regime characteristics influence the lethality of incidents?

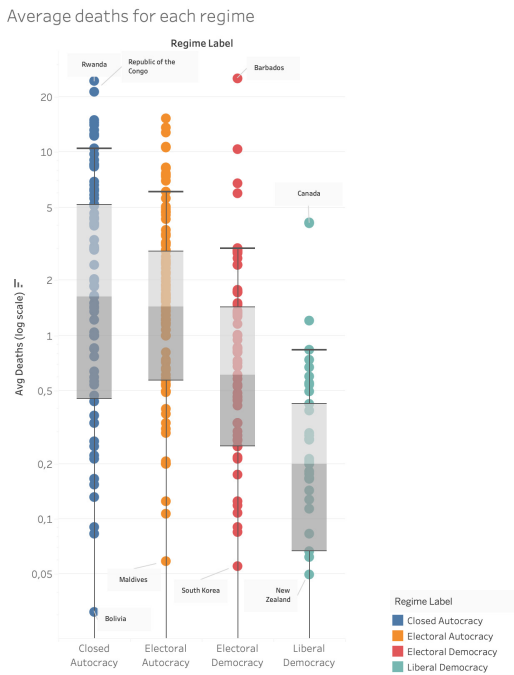


Figure 54: Average deaths for each regime per country

This visualisation takes the form of a strip plot with interquartile range overlays, plotting average deaths per country-year on a logarithmic scale across the four regime categories. The logarithmic scaling is necessary given the extreme dispersion of the data, with outliers such as Rwanda and the Republic of the Congo sitting well above the bulk of Closed Autocracy observations, and Barbados appearing as an outlier within Electoral Democracies. The median lines and interquartile boxes allow for a systematic comparison of central tendency and spread across regime types. The substantive picture that emerges is one where Closed Autocracies display the highest median lethality and the greatest internal variation, followed by Electoral Autocracies, with Electoral and Liberal Democracies occupying progressively lower and more compressed positions. This pattern suggests that when attacks occur in fully authoritarian contexts, they tend to be considerably more deadly on a per-incident basis, even if their aggregate frequency is lower. The finding points to an important distinction between the volume and the severity of political violence across regime types.

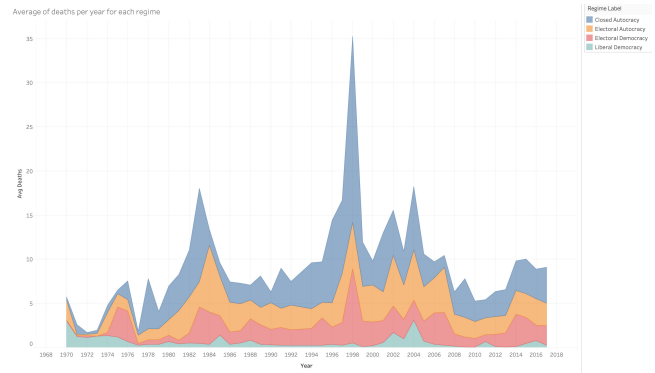


Figure 55: Average of deaths per year for each regime broken down by years

This stacked area chart tracks the average number of deaths per terrorist incident annually from 1970 to approximately 2020, disaggregated by regime type. Unlike the frequency chart presented in the previous section, which was dominated by Electoral Autocracies during the 2010s, the lethality chart tells a markedly different story, one in which Closed Autocracies consistently account for the largest share of average deaths across most of the time series, particularly during the pronounced spikes of the mid-to-late 1990s. The peak around 1994–1998 is especially revealing, reaching over 35 average deaths per incident and dwarfing any other period in the dataset. This almost certainly reflects mass-casualty events concentrated in fully closed or near-closed authoritarian contexts, most notably the Rwandan genocide and related political violence in Central Africa. This temporal decoupling between frequency and lethality is analytically significant: while hybrid regimes generate the most attacks in absolute terms, closed autocracies appear to be the environment in which individual incidents produce the most devastating human costs, suggesting that the near-total absence of accountability mechanisms and civil society oversight enables violence of a qualitatively different and more catastrophic nature.

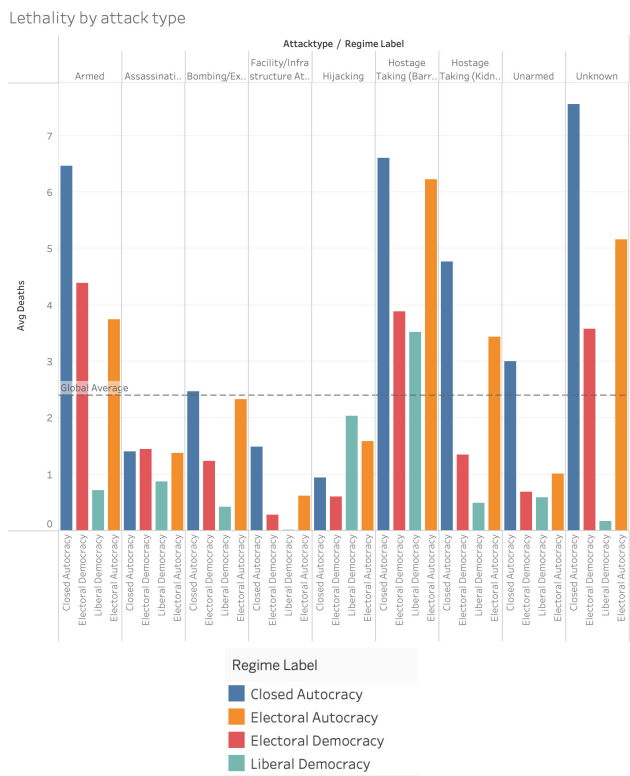


Figure 56: Lethality by attack type and by regime

The grouped bar chart disaggregates average deaths per incident by attack type, with bars for each regime type presented side by side within each attack category and a dashed global average reference line running horizontally across the chart. This design allows for the simultaneous comparison of how lethality varies both across methods of attack and across political contexts. Across nearly all attack categories (armed assault, assassination, facility attacks, and hostage-taking), Closed Autocracies consistently record the highest average deaths per incident, frequently exceeding the global average. Electoral Autocracies perform comparably in several categories, particularly in hostage-taking and unknown attack types. Liberal Democracies, by contrast, sit at or below the global average across virtually all attack types. The pattern holds remarkably consistently regardless of the method used, suggesting that the institutional characteristics of the regime, rather than the tactical choices of perpetrators, are a primary determinant of how lethal any given attack turns out to be.



Figure 57: Lethality and Regime by Country - 2017

The final visualisation in this section is a bubble map for the year 2017, where bubble size encodes average deaths per incident and colour identifies the regime type of each country. Unlike the frequency map presented earlier, which highlighted Iraq, Afghanistan, and Pakistan as the most affected countries, the lethality map shifts attention toward a somewhat different set of countries, with several large bubbles appearing across Sub-Saharan Africa, the Middle East, and parts of South Asia. Notably, several of the largest bubbles in terms of average deaths per incident correspond to Closed Autocracies and Electoral Autocracies in Africa, reinforcing the finding from the temporal chart that the most lethal individual attacks tend to be concentrated in the most institutionally constrained environments. The geographic distribution also reveals that high lethality is not exclusively a feature of countries with high attack frequency, since some countries with comparatively few events record disproportionately deadly incidents, further underlining the analytical value of treating frequency and lethality as distinct dimensions of terrorist violence shaped by different facets of political regime type.

(3) Is there a relationship between political stability and attack success rates?

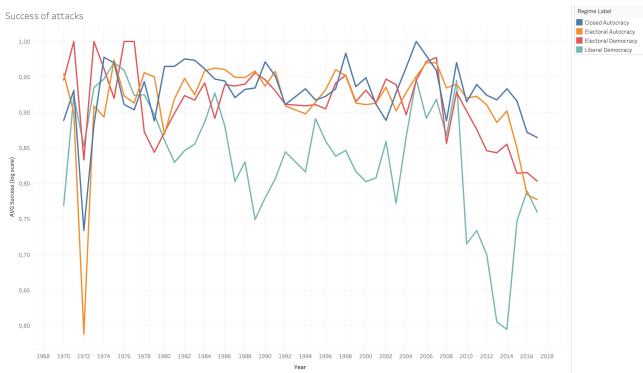


Figure 58: Success of attacks by year and regime

The line chart tracks average annual attack success rates from 1970 to approximately 2018, comparing the four regime types on a logarithmic scale. Throughout most of the period, autocracies (dark blue and orange lines) maintain consistently higher average success

rates than democracies (red and teal lines), hovering between 0.90 and 0.97 for most of the series. Liberal democracies (teal line) stand out for their greater volatility and persistently lower values across the entire period, dropping below 0.60 in the early 1970s and again around 2014. From approximately 2008 onward, all four categories register a sharper and increasingly convergent decline. Two features of this chart deserve particular attention. First, the persistent hierarchy between autocratic and democratic regimes across five decades suggests a structural rather than incidental relationship between regime type and attack success rates, reinforcing patterns identified in other visualisations. The weaker institutional capacity and lower security responsiveness typically associated with autocratic settings appear to create more permissive conditions for the operational effectiveness of terrorist groups. Second, the broad convergence and shared decline in the final decade may reflect the global expansion of counterterrorism cooperation, intelligence sharing, and security capacity building in the post-9/11 period, which progressively eroded the operational advantage terrorist groups had historically enjoyed in weaker institutional environments. The overall trajectory thus captures both a durable regime-type effect and a more recent systemic shift in the global security landscape.

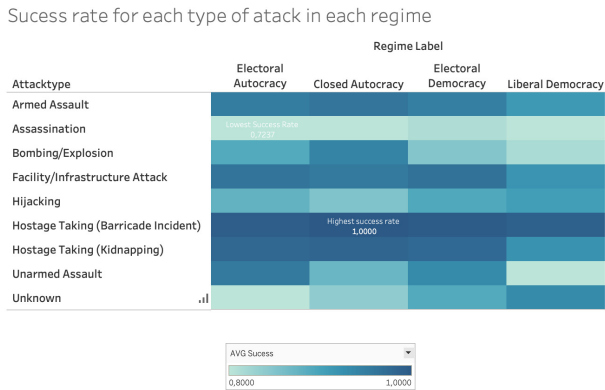


Figure 59: Success rate for each type of attack in each regime

The heat map displays average attack success rates across nine attack types and four regime categories, with colour intensity encoding the success rate on a scale from approximately 0.80 to 1.00. The annotation highlights the lowest observed success rate of 0.7237, recorded for armed assassination in Electoral Autocracies, and the highest of 1.0000, recorded for hostage-taking barricade incidents in Closed Autocracies. The relatively compressed colour range, with most cells sitting between 0.85 and 1.00, indicates that attack success rates are generally high across all regime types, leaving limited but analytically meaningful variation to interpret. Despite this compression, some patterns are discernible. Closed Autocracies tend toward darker shades across several attack types, suggesting marginally higher success rates in those contexts, which may reflect weaker or more selectively deployed security responses. Liberal Democracies, by contrast, display lighter shading in several categories, consistent with the expectation that more developed institutional and security infrastructures reduce the likelihood of an attack achieving its intended objective. The heat map thus offers initial evidence that

regime type shapes not only how often and how lethally attacks occur, but also how frequently they succeed.

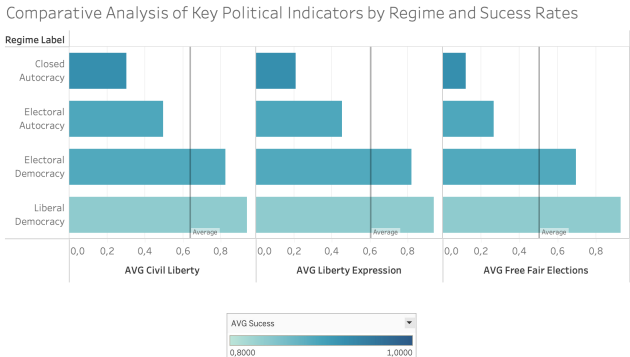


Figure 60: Success rate for each regime in comparative indicators

This grouped horizontal bar chart plots three institutional indicators (average civil liberty, average liberty of expression, and average free and fair elections) across the four regime types, with bar colour encoding average attack success rates. The chart is designed to connect the structural characteristics of political regimes to the operational outcomes of terrorist attacks, allowing for a direct visual assessment of whether more open and rights-protective institutions are associated with lower attack success. The pattern that emerges is intuitive but important: Liberal Democracies score highest on all three indicators and display the lightest bar colour, suggesting lower average attack success rates. Closed Autocracies score lowest and show darker shading, indicating higher success rates. This inverse relationship between institutional openness and attack success is consistent with the argument that democratic regimes, despite being more permeable targets in some respects, invest more heavily in security infrastructure and rule-of-law mechanisms that ultimately reduce the operational effectiveness of terrorist groups.

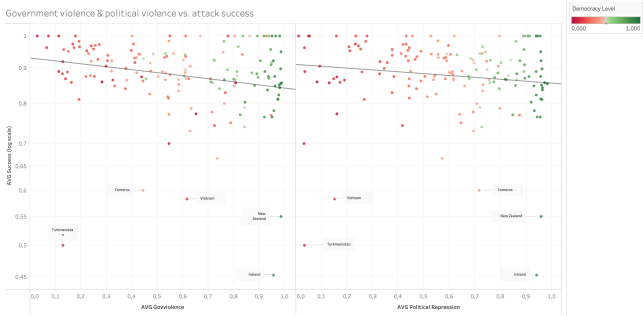


Figure 61: Government violence and political violence Vs. attack success in comparison with democracy levels

This dual scatter plot places average government violence (left panel) and average political repression (right panel) on the x-axes, with average attack success on the y-axis on a logarithmic scale. Each point represents a country, coloured by democracy level on

a red-to-green gradient, allowing simultaneous reading of repression intensity, regime openness, and operational terrorist outcomes. Trend lines with a slight negative slope are fitted across both panels. Both panels tell a broadly consistent story: higher levels of government violence and political repression are associated with marginally lower attack success rates, as captured by the downward-sloping trend lines. Countries such as Turkmenistan and Comoros, which combine high repression with low democracy scores (darker red), appear as outliers with notably low success rates, suggesting that highly coercive states may be particularly effective at disrupting or neutralising attacks before they achieve their objectives. Conversely, Ireland and New Zealand, high-democracy, low-repression states, appear at the lower end of the success axis, which appears counter-intuitive but likely reflects the robustness of their institutional security frameworks rather than repressive capacity. Overall, the relationship is modest and dispersed, indicating that repression alone is not a determinative factor in attack success.

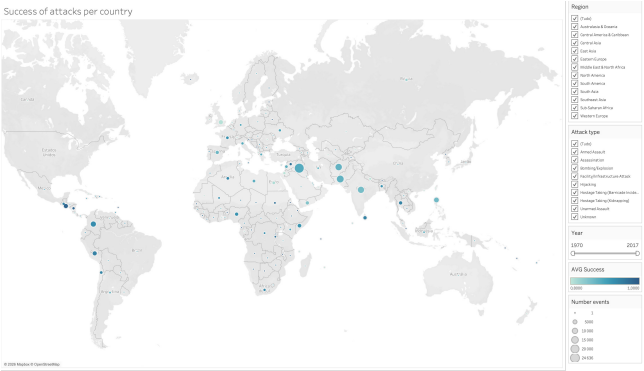


Figure 62: Success of attacks per country

This bubble map displays average attack success rates geographically, with bubble size encoding the total number of events and colour encoding average success on a light-to-dark teal scale. The full range of years (1970–2017) and attack types is included, with interactive filters for region, attack type, and year visible on the right panel. The geographic distribution of success rates reveals no single dominant regional pattern, but several observations stand out. Countries across Sub-Saharan Africa, the Middle East, and parts of South and Southeast Asia and Central/South America display larger bubbles, reflecting higher event volumes, with generally darker colouring, suggesting both high frequency and high success rates. Western Europe and North America, while not entirely absent, tend toward smaller and lighter markers. This spatial pattern is consistent with the broader argument that institutionally fragile or conflict-affected states provide more permissive environments for the operational success of terrorist attacks, independent of any single regime characteristic.

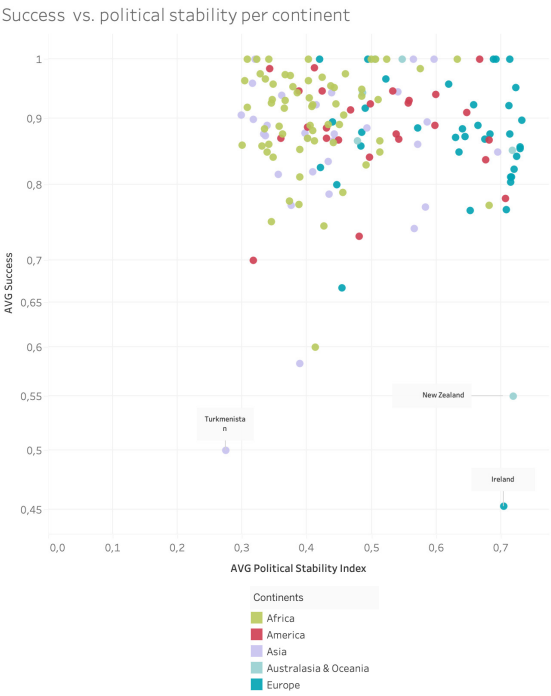


Figure 63: Success and political stability per continent

This scatter plot maps average political stability index scores on the x-axis against average attack success on the y-axis, with each point representing a country coloured by continent. The distribution is notably non-linear: most countries cluster in the upper-middle range of the chart regardless of their stability score, with success rates concentrated between 0.80 and 1.00 across virtually the full stability spectrum. The absence of a clear linear relationship is itself substantively informative. Countries such as Ireland and New Zealand, which have relatively high stability scores, nonetheless record some of the lowest success rates in the dataset, likely due to the strength of their security and judicial institutions rather than political stability. Turkmenistan, by contrast, combines moderate stability with a low success rate, reflecting its highly coercive state apparatus. The scatter thus suggests that political stability and attack success are only weakly correlated in the aggregate, and that the relationship is better understood through the lens of specific institutional characteristics (rule of law, security capacity, and civil liberties) rather than stability as a composite index.

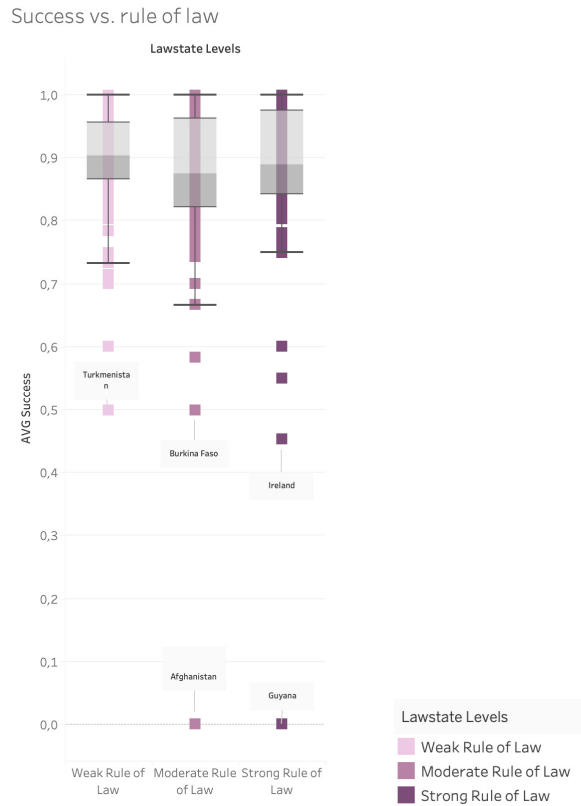


Figure 64: Success VS. Rule of law for each country

The final visualisation categorises countries into three rule-of-law levels (weak, moderate, and strong) and plots the distribution of average attack success rates for each category using a square mark strip plot with interquartile overlays. Outliers are labelled, including Turkmenistan, Afghanistan and Burkina Faso at the lower end of the weak and moderate categories respectively, and Ireland and Guyana as low-success outliers within the strong and moderate categories. The central finding is that median success rates are remarkably similar across all three rule-of-law categories, with interquartile boxes overlapping substantially. However, the tails of the distribution differ in meaningful ways: the weak rule-of-law category displays greater downward dispersion, suggesting that while most attacks succeed regardless of legal institutions, the most extreme failures tend to cluster in settings where the state is weakest. This counter-intuitive pattern may reflect that in very weak institutional environments, both the security apparatus and the terrorist groups themselves operate with high unpredictability. Strong rule-of-law states, while displaying a similarly high median, show a tighter distribution, consistent with more systematic and effective, if not dramatically more successful in outcome terms, counterterrorism responses.

7 Proposal

The final dashboards were designed based on the research questions outlined in the previous section. A total of eight interactive

dashboards were developed, each addressing a distinct analytical perspective.

7.1 Dashboard 1: Lifecycle of Terrorist Groups: Activity and Geography

This dashboard focuses on characterizing the temporal and geographic dimensions of the most active terrorist groups. The main visualisation, Lifecycle by group, displays the frequency of events over time for each organisation, allowing users to identify when each group emerged, peaked, and declined. The analysis is complemented by key performance indicators (KPIs) that highlight crucial milestones, such as the first year, last year, and peak year of activity, alongside averages and total event counts. Rather than cluttering the main view, geographic distribution and deep-dive data are integrated into interactive tooltips. When hovering over a data point, the tooltip reveals detailed insights, including the group name, specific year, event counts and a breakdown of attack regions. To support exploratory analysis, users can dynamically filter the data using the controls on the top panel, which include a year range slider, a region dropdown and a total events slider.

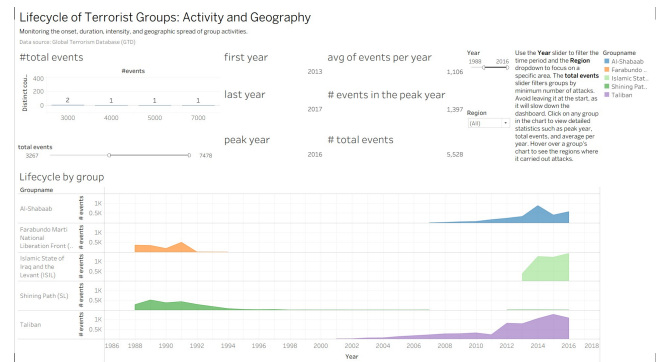


Figure 65: Lifecycle of Terrorist Groups: Activity and Geography

7.2 Dashboard 2: Tactical Profile

This dashboard focuses on the behavioural and operational profile of each terrorist group, addressing questions about attack methods and weapon preferences. It combines three visualisations: a heatmap of attack types by group, showing each group's tactical preferences, a heatmap of lethality by weapon subtype and group, capturing how deadly each weapon category tends to be for each group, and a stacked bar chart showing the frequency of attacks by group and weapon subtype, offering a direct comparison of the most commonly used weapons across groups. Two sliders, one for the number of top groups and one for the number of top weapons, allow users to focus on the most relevant actors and arsenals. To further support exploratory analysis, tooltips are available throughout the dashboard, providing on-demand detail for individual data points, including group name, attack type, number of events, weapon type, weapon subtype and percentage of lethality, without cluttering the main visualisations.



Figure 66: Tactical Profile

7.3 Dashboard 3: Global Terrorism Landscape: Scale, Distribution, and Mortality

This dashboard provides a broad overview of global terrorism patterns, serving as the contextual entry point for the analysis. It includes a line chart of total attack frequency over time, a choropleth world map showing the geographic distribution of attacks by country, a bar chart of attack frequency by region, and a bar chart of total deaths by region. A dropdown filter for region and a year range slider allow users to explore how the global landscape of terrorism has shifted across different periods and geographic areas. This dashboard highlights that terrorism is not uniformly distributed, the Middle East and North Africa and South Asia account for the vast majority of both attacks and fatalities, and that the overall frequency of incidents increased dramatically from the mid-2000s onwards, peaking around 2014. To further support exploratory analysis, tooltips are available throughout the dashboard, providing on-demand detail for individual data points, including year, number of events, region and top group names in each country, without cluttering the main visualisations.

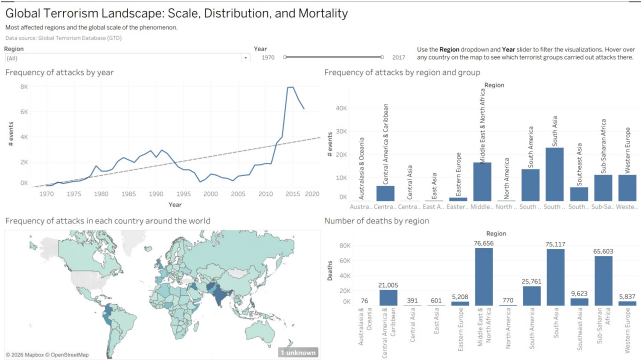


Figure 67: Global Terrorism Landscape: Scale, Distribution, and Mortality

7.4 Dashboard 4: Attack Dynamics and Impact: Lethality, Success, and Victims Over Time

This dashboard addresses the second research question by examining how attack type, target type, and region jointly influence the lethality and success of terrorist attacks over time. It includes a stacked bar chart on a log scale showing the relationship between attack type and target type in terms of number of deaths, a bar chart of attack success rate by region, a line chart of average lethality over time, and an area chart showing the total number of victims alongside the number of events per year. Filters for target type and year range allow for flexible exploration of these relationships. Key findings visible in this dashboard include that armed assaults and bombings are responsible for the highest death tolls, that success rates are uniformly high across all regions (above 84%), and that the surge in victims after 2010 is driven more by an increase in the number of events than by a rise in per-attack lethality. To further support exploratory analysis, tooltips are available throughout the dashboard, providing on-demand detail for individual data points, including attack type, target, number of deaths, number of injured, number of events, region, percentage of attacks success, year and average lethality, without cluttering the main visualisations.

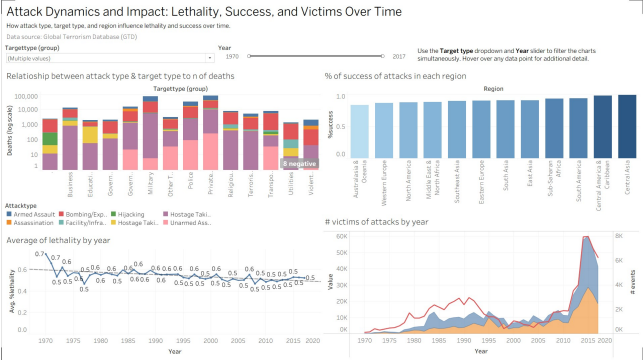


Figure 68: Attack Dynamics and Impact: Lethality, Success, and Victims Over Time

7.5 Dashboard 5: Political regimes & frequencies of attacks

The visual evidence in this dashboard points to a clear answer that the political regime type is strongly associated with the frequency of terrorist attacks, but the relationship is more complex than a simple hierarchy. The area chart shows that until the early 2000s, attacks were relatively evenly distributed across regimes. From that point onward, and dramatically until 2014, it is Electoral Autocracies and Closed Autocracies that dominate the growth, driven by events such as the post-9/11 period, the Arab Spring, and the ISIS peak. Liberal democracies maintain a consistently small share throughout the entire time series, suggesting that consolidated democratic institutions act as a protective factor. The scatter plot reinforces this reading in a nuanced way: countries with intermediate democracy values (between 0.1 and 0.4), typically electoral autocracies, concentrate the highest density of attacks and the largest absolute death

tolls (Iraq, Afghanistan, Pakistan). Notably, both very closed and very open regimes show lower frequencies, suggesting a non-linear relationship: peak risk lies in the space of political transition, where the state exists but is weak or illegitimate. The 1997 map illustrates this geographic pattern well even in a year of moderate activity.

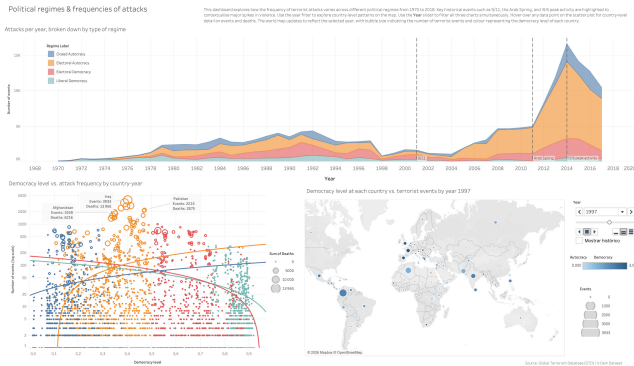


Figure 69: Political regimes & Frequencies of attacks

7.6 Dashboard 6: Terrorism & Regime Type – Lethality Analysis

The answer conveyed by this dashboard is that regime characteristics influence the lethality of incidents, but in a counterintuitive way and depending on the attack method. The box plot of average deaths per regime reveals that Closed Autocracies show the greatest dispersion and most extreme outliers (Rwanda, Republic of the Congo), indicating that when violence occurs in these contexts it can be devastating. Liberal Democracies show the most compressed and lowest distribution, with rare but occasionally lethal attacks. Electoral Democracies sit in between with high variance. The temporal chart of average lethality confirms that the 1990s were the peak of lethality per attack, not the 2010s, which saw more events but lower average lethality per incident. This suggests that the surge in ISIS-related volume was accompanied by individually less deadly attacks on average. The bar chart by attack type is particularly revealing: Armed Assaults in Closed Autocracies and Hostage Taking in Electoral Autocracies far exceed the global average, while in Liberal Democracies nearly all attack types fall below that line. Regime type does not only determine how many attacks occur, but it also shapes the type of violence deployed and its lethal capacity.

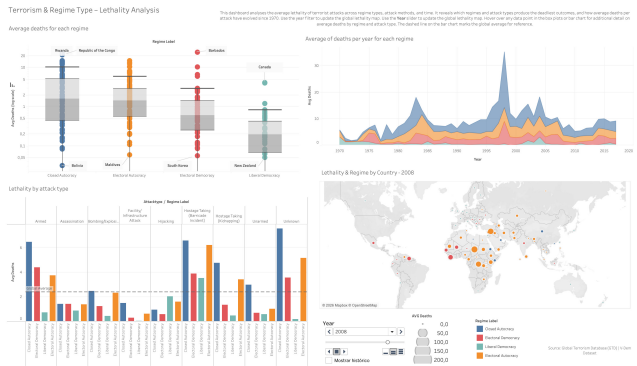


Figure 70: Terrorism & Regime Type – Lethality Analysis

7.7 Dashboard 7: Terrorism & Success of Attacks by regime

This dashboard studies the relationship between political stability and attack success rates through the lens of regime type and governance indicators, revealing that the relationship between political stability and attack success is real but surprisingly subtle. The time series of average success rates shows that all four regime types maintain consistently high success rates, rarely dropping below 0.8 throughout the entire 1970–2018 period. This is a striking finding in itself, regardless of regime, the vast majority of terrorist attacks succeed. However, meaningful differences emerge in the trends. Liberal Democracies show the steepest decline from the 2000s onward, reaching the lowest values by the end of the period, which likely reflects sustained investment in counter-terrorism infrastructure and intelligence capabilities in these more institutionally stable societies. Closed and Electoral Autocracies, by contrast, remain stubbornly high throughout. The heatmap of success rates by attack type and regime adds important nuance. Hostage Taking (Barricade Incident) in Electoral Autocracies achieves a perfect 1.0 success rate, suggesting complete state incapacity or unwillingness to intervene, while Assassinations in Liberal Democracies record the lowest success rate in the entire matrix (0.7237), precisely the attack type where state protection of individuals is most directly tested. This contrast is telling that stable democracies are better at protecting specific targets but struggle to prevent attacks altogether. The comparative bar chart of political indicators reinforces this narrative. Liberal Democracies score highest on civil liberties, freedom of expression, and free and fair elections, and also show the lowest average success rates. This alignment suggests that institutional quality and political openness, rather than repression, are the mechanisms through which stability reduces attack effectiveness.

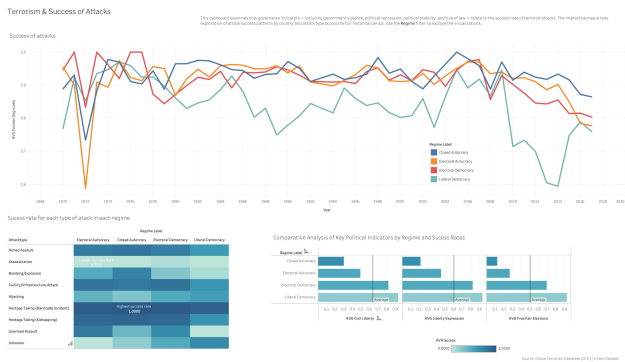


Figure 71: Terrorism & Success of Attacks by regime

7.8 Dashboard 8: Terrorism Success & Institutional Quality

This dashboard zooms into specific institutional indicators, offering a more granular and mechanistic reading of the same question. The scatter plots of government violence and political repression against attack success reveal a weak but discernible negative relationship: as democracy level increases (greener dots), success rates tend to decline slightly. However, the dispersion is wide and several outliers challenge any simple interpretation. Ireland and New Zealand, both highly democratic and stable, appear with notably low success rates, consistent with the hypothesis that strong institutions constrain terrorist effectiveness. Turkmenistan, on the other hand, sits at the opposite extreme with very low success despite high repression, which may reflect underreporting, distinct conflict dynamics, or the near-absence of organised terrorism in heavily controlled states. In the Success vs. Political Stability Index scatter plot, countries at the extremes of the stability spectrum, either very unstable or very stable, tend to show lower attack success rates, while those in the intermediate range experience the highest. Very unstable states may lack the organisational capacity even to sustain consistent terrorism, while very stable ones invest heavily in prevention. It is the fragile middle ground (states with enough structure to be targeted but not enough resilience to resist) where attacks succeed most reliably. The Success vs. Rule of Law box plots further support this: countries under weak rule of law show greater variance and some extreme low outliers, while strong rule of law contexts compress outcomes downward. The world map of attack success by country visually confirms the concentration of high success rates across Central Asia, Sub-Saharan Africa, Central/South America, and parts of the Middle East, precisely the regions where institutional quality and political stability are most fragile.

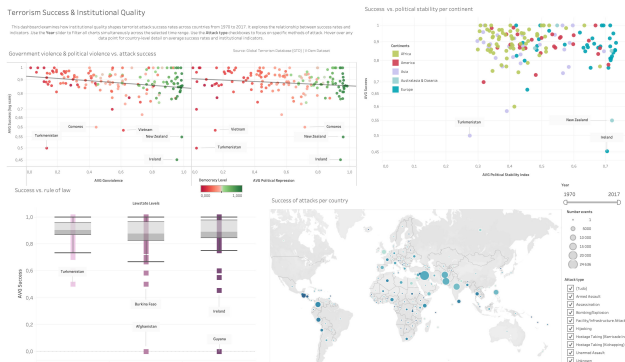


Figure 72: Terrorism Success & Institutional Quality

8 Conclusion

This work set out to explore the dynamics of global terrorism through a data-driven approach, combining the Global Terrorism Database (GTD) with the V-Dem political indicators dataset, and translating the resulting analysis into eight interactive dashboards built in Tableau. Regarding the first research question, the lifecycle of terrorist groups, the analysis revealed that terrorist organisations follow distinct temporal patterns of emergence, escalation, and decline, with a clear geographic anchor for most groups. The post-2010 period is dominated by a small number of extraordinarily prolific organisations, most notably ISIL, Boko Haram, and the Taliban, which together account for a disproportionate share of all attacks and fatalities in the dataset. Bombing and explosion consistently emerges as the dominant attack method across groups, while weapon preferences and target selections reflect each group’s operational context and strategic objectives. These findings are addressed in Dashboards 1 and 2.

The second research question, how attack type, target type, and region jointly influence lethality and success, produced several counter-intuitive results. While bombing is the most frequent attack method, hostage-taking events tend to produce the highest average deaths per incident. Private citizens and property constitute the most targeted category by a wide margin, reflecting a strategic shift toward maximising civilian casualties. Attack success rates are uniformly high across all regions, consistently above 84%, suggesting that prevention at the point of execution remains a persistent challenge regardless of geographic or institutional context. These patterns are explored in Dashboards 3 and 4.

The third research question, how political regimes shape the frequency, lethality, and success of terrorist attacks, yielded the most analytically rich findings. Electoral autocracies are systematically associated with the highest attack frequencies, while closed autocracies, despite lower event counts, produce the most lethal individual incidents. Liberal democracies consistently register the lowest frequency, lowest lethality, and lowest attack success rates across the entire time series, suggesting that consolidated democratic institutions act as a structural protective factor. Crucially, however, the relationship between regime type and terrorism is non-linear: the greatest risk is concentrated in states undergoing political transition, where institutional capacity is weak or contested.

These findings, visualised in Dashboards 5 through 8, directly address the gaps identified in the state of the art, namely the absence of interactive tools connecting V-Dem indicators with GTD attack data.

In summary, this project demonstrates that terrorism is not a random phenomenon but one deeply structured by political context, geographic concentration, and organisational behaviour. The integration of political regime indicators with terrorism data into an exploratory visual platform constitutes the central original contribution of this work, enabling intuitive and interactive exploration of relationships that existing studies have addressed only through static statistical methods.

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